

Slack Induction by String Removals for Vehicle Routing Problems

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Dedicated algorithm and modelling improvements continue to advance the state-of-the-art with respect to solving vehicle routing problems (VRPs). Despite these academic achievements, solving large VRP instances sufficiently fast for real-life applicability remains challenging. This paper therefore advocates the employment of a fast optimization heuristic which exploits a VRP solution's characteristics in an effective manner. This work's primary contributions are threefold: a ruin method, a recreate method and a fleet minimization procedure. The ruin method functions via adjacent string removal, introducing with it a novel property regarding vehicle routing problems which we term spatial slack, while the recreate method is categorised as greedy insertion with blinks. Combining these results in SISRs: a powerful ruin & recreate approach. The fleet minimization procedure, meanwhile, introduces an absences-based acceptance criterion which serves as a complementary optimization component for when minimizing the number of vehicles constitutes the primary VRP objective. Together, these three elements provide a suite of simple, powerful and easily-reproducible algorithmic methods which are not only successfully applied to the CVRP but also for a wide range of related problems such as pickup and delivery problems as well as others which include time windows. SISRs serves to strip back the layers of complexity and specialisation synonymous with the trend of algorithmic development throughout recent decades. Moreover, such simplicity and reproducibility are shown to not necessarily come at the expense of solution quality, with SISRs consistently outperforming alternative general approaches as well as dedicated single-purpose methods, obtaining highly competitive results and new best results, particularly for large-scale instances. Finally, aside from performance-related criteria, SISRs also serves to showcase a fresh perspective with respect to vehicle routing problems more generally, introducing a range of new terminology and procedures which it is hoped will invigorate further research and innovation.

Key words: vehicle routing problem, large-scale instances, string removal, spatial slack, capacity slack, fleet minimization, ruin & recreate

1. Introduction

The capacitated vehicle routing problem (CVRP) was introduced by Dantzig and Ramser (1959) as the truck dispatching problem, a generalization of the traveling salesman problem (TSP). Despite multiple decades of research (Laporte 2009), the CVRP remains a highly active field of research and it continues to represent a significant computational challenge. Research concerning

the problem mostly falls into one of two categories: exact algorithms or heuristics.

Developing exact algorithms is an interesting activity in itself while also providing objective function bounds which enable accurate heuristic quality assessment. Toth and Vigo (2002) provide a comprehensive overview of exact algorithms to solve CVRPs with either symmetric or asymmetric distance matrices. They primarily focus on branch-and-bound algorithms and review a set of original powerful approaches. The branch-cut-and-price (BCP) algorithm by Fukasawa et al. (2006), improved upon previous branch-and-cut algorithms. Several improvements such as by Baldacci, Christofides, and Mingozzi (2008) are combined with new elements (Pecin et al. 2014), and prove capable of solving CVRPs of up to 350 customers and even 650 in some cases. Computation times may, however, be as long as five days, which makes such solvers impractical for many real-world purposes.

Heuristics, by contrast, are characterized as fast and adaptable methods. Absolute optimality is, however, not ensured - a necessary trade-off in terms of accommodating many real-world situations which rely on the presence of timely solutions. Heuristics generally employ specific neighborhoods which are employed by a metaheuristic to gradually improve candidate solutions. Prior to the latest improvements, classical neighborhoods such as 2-opt* (Potvin and Rousseau 1995) and CROSS-exchange (Taillard et al. 1997) are explored in single-solution-based heuristics. Larger neighborhoods are employed by Shaw (1998) and Pisinger and Røpke (2007) in a ruin & recreate framework. Prins (2004) contributed a hybrid genetic algorithm, the first population-based framework for VRPs, which improved significantly upon the prior approaches. Recently this approach was refined by Vidal et al. (2015), thereby realizing the state-of-the-art VRP heuristic.

The vast majority of VRPs consider minimizing total travel distance as their objective. However, a much smaller number of variants require a hierarchical objective to be handled which is primarily the minimisation of the number of vehicles and secondarily the minimisation of the total distance. In such problems, solutions associated with smaller fleet sizes are preferred over solutions whose total travel distance is smaller, but which require additional vehicles. Although this objective might appear artificial, it reflects the attitude of real-world transportation companies who far prefer to be able to accommodate all requests with their own fleet, rather than having to outsource certain requests to other companies - the cost of which is prohibitively high. Solution methods which address this hierarchical objective are generally two-stage approaches wherein the number of vehicles is minimised in the first stage and the distance in the second stage, as the methods by Pisinger and Røpke (2007), Nagata, Bräysy, and Dullaert (2010). Hybrid methods, for example by

(Vidal et al. 2013), are less common but also possible.

While the initial ambition of heuristics is noble, they have gradually become increasingly complicated. The present paper seeks to remedy this by introducing a low-level, yet simultaneously powerful and fast, approach which is sufficiently adaptable and, as such, may be easily incorporated into any current or future VRP approach. This ruin & recreate approach's improvements to CVRP state-of-the-art and its, not only successful but also competitive, application to a host of related vehicle routing problems, as will subsequently be demonstrated and documented.

Structurally, the paper begins by first offering a detailed problem definition of the CVRP, followed by an overview of the other vehicle routing problems addressed during experimentation. Next, Section 3 provides an introduction to ruin & recreate while Section 4 provides a brief description of Simulated Annealing, the metaheuristic employed for guiding the local search. Section 5 introduces SISRs, *Slack Induction by String Removals*, this paper's novel ruin & recreate contribution. Following this, the new *absences-based acceptance criterion* for fleet minimisation is detailed in Section 6. Parameters associated with this work are detailed in Section 7. Computational results for the CVRP and a variety of associated problems are provided in the following section and the paper ultimately ends with a conclusion which summarises the paper's contribution while also delineating the scope and possibilities for future research.

2. Problem Description

The CVRP, considered by the current paper, is defined as follows. Let $\mathcal{G} = \{\mathcal{V}, \mathcal{A}\}$ be a complete undirected graph in which $\mathcal{V}$ is the set of vertices and $\mathcal{A}$ the set of arcs connecting the vertices. The vertices $v_i \in \mathcal{V}$ for $i \in \{0, \dots, n\}$ represent locations in a 2-dimensional space where v_0 corresponds to the *depot* and the other n vertices with *customers* having a demand q_i . Each arc (i, j) in $\mathcal{A} = \{(i, j) : i, j \in \mathcal{V}, i \neq j\}$ is associated with a *cost* c_{ij} . An unlimited homogeneous fleet of vehicles with capacity Q is situated at the depot. The CVRP consists of designing vehicle *tours* at minimum cost such that each customer is served exactly once by a single vehicle while each tour starts at the depot v_0 , serves customers without exceeding the vehicle's capacity Q and finally ends at the same depot v_0 .

Table 1 CVRP and associated problems.**CVRP (Capacitated Vehicle Routing Problem)**

The Capacitated VRP, also known as the classical VRP, was introduced by Dantzig and Ramser (1959) as the truck dispatching problem. The algorithms by Subramanian, Uchoa, and Ochi (2013) and Vidal et al. (2014) are considered as the current best approaches.

MDVRP (Multi-Depot VRP)

Vehicles are available at a fixed number of depots. Vehicles are assumed to return to the depot where their tour starts. This problem was introduced by Gillett and Johnson (1976). Pisinger and R pke (2007), Subramanian, Uchoa, and Ochi (2013) and Vidal et al. (2012) are considered as the best solution approaches.

VRPB (VRP with Backhauls)

The vehicle routing problem with Backhauls, introduced by Goetschalckx and Jacobs-Blecha (1989), considers serving two sets of customers: delivery (linehaul) and pickup (backhaul) customers. Each tour must serve at least one delivery customer and all delivery customers must be served before any pickup customers are served. A tour's total delivery and pickup demand should not exceed its vehicle's capacity. The algorithms by Gajpal and Abad (2009), Zachariadis and Kiranoudis (2011) and Vidal et al. (2014) represent the best solution approaches for this problem.

VRPMDP (VRP with Mixed Delivery & Pickup)

This problem, introduced by Min (1989), relaxes the VRPB by enabling any order of delivery and pickup customers within a tour. This implies that a vehicle's used capacity may increase and decrease during the tour, but should never exceed the vehicle's capacity. Best solutions are obtained by Subramanian, Uchoa, and Ochi (2013) and Vidal et al. (2014).

VRPSDP (VRP with Simultaneous Delivery)

Unlike the VRPMDP this problem, also introduced by Min (1989), does not require customers to be associated exclusively with either delivery or pickup demand. This implies that a single customer can have both delivery and pickup requests. Best solutions are obtained by the same authors as for the VRPMDP.

MDVRPMDP (Multi-Depot VRP with Mixed Delivery & Pickup)

Min (1989) generalizes the VRPSDP by addition of multiple depots into the Multi-Depot VRPSDP. Best approaches are those of R pke and Pisinger (2006b) and Subramanian, Uchoa, and Ochi (2013).

OVRP (Open VRP)

In contrast to the CVRP, vehicles must not return to the depot after serving the last customer in the Open VRP, introduced by Repoussis et al. (2010). This reflects situations in which customers are assigned to external carriers. Current best approaches are those of Subramanian, Uchoa, and Ochi (2013) and Vidal et al. (2014).

CCVRP (Cumulative CVRP)

The Cumulative CVRP, introduced by Nogueu, Prins, and Wolfer Calvo (2010), requires the sum of arrival times at customers to be minimized rather than the total traveled distance as required by the CVRP. Vidal et al. (2014) obtained the current best known solutions for this problem.

VRPTW (Vehicle Routing Problem with Time Windows)

Gehring and Homberger (1999) generalized the CVRP with temporal constraints. Customers are associated with individual time windows wherein they prefer to be served. Vehicles start their tour in the depot and are assumed to arrive back at the depot without exceeding the maximum tour duration. The current best approaches are those of Repoussis, Tarantilis, and Ioannou (2009), Nagata, Br ysy, and Dullaert (2010) and Vidal et al. (2013).

PDPTW (Pickup & Delivery Problem with Time Windows)

In the Pickup & Delivery problem, introduced by Dumas, Desrosiers, and Soumis (1991), goods are transported between pickup and delivery nodes. The PDPTW should not be mixed with Delivery & Pickup problems where goods are transported from or to the depot, not between the customers themselves. In addition to these coupling constraints, similar temporal constraints are present as in the VRPTW. R pke and Pisinger (2006a), Nagata and Kobayashi (2010) and Curtois et al. (2018) contributed the current best solution approaches.

While the CVRP constitutes the primary problem addressed by the present paper, the general applicability of SISRs is demonstrated by way of solving several other associated problems within the domain of vehicle routing. Key references are supplied for papers which provide a description for each individual problem in Table 1.

The remainder of this paper refers to a *string* as a sequence of consecutive nodes in a tour. Strings may contain at most all customers served by the tour. The number of customers included in a string is referred to as the *cardinality* of the string. The number of customers served by a tour t is referred to as the tour's cardinality and denoted by $|t|$. Solution s is represented by two sets $s = \{T, A\}$. T represents the set of all tours and A represents the set of *absent customers* which are not served by any tour $t \in T$. Therefore, a solution s may only be feasible if A is an empty set.

3. Ruin & Recreate Strategies

Classical neighborhoods for VRP heuristics - such as, 2-opt (Croes 1958), 2-opt* (Potvin and Rousseau 1995), k -opt (Lin 1965), Or-opt (Or 1976), exchange, cross and relocate (Savelsbergh 1988) and CROSS-exchange (Taillard et al. 1997) - are all defined via slight modifications to the incumbent solution. They therefore exhibit limited neighborhood size complexity, implying a full neighborhood $\mathcal{N}(s)$ of a solution s may be evaluated within a reasonable amount of computational time. Very large neighborhoods often occur when a significant part of the solution is modified, making a full neighborhood exploration impractical. Therefore, neighbors may be selected from a reduced candidate set $\mathcal{C}(s) \subseteq \mathcal{N}(s)$ or a single neighbor may be generated by first ruining the current solution before recreating it into a feasible one. Such a method may be defined by a *ruin method* $\mathcal{R}^-$ and *recreate method* $\mathcal{R}^+$. This underlying concept is present in a variety of research papers. To our knowledge, it was established for the first time by Dees and Smith II (1981) in their *Rip-Up and Reroute* strategies for wiring point-to-point connections in electronic design automation. Shaw (1998) introduced *Large Neighborhood Search* (LNS) wherein the ruin phase is implemented as the removal of related customers (related in terms of time, distance and being served by the same vehicle). A branch-and-bound technique optimally inserts the removed customers in the recreate phase. Shaw (1998) applied LNS to both the CVRP and VRPTW. The term *Ruin & Recreate* (R&R) was introduced by Schrimpf et al. (2000) who applied the technique to a number of prominent problems including the TSP and VRPTW. They ruined solutions by removing either randomly selected customers, customers within a certain radius or one single string. The solution is recreated by greedily reinserting the removed customers in a random order at minimum cost. R pke and Pisinger (2006a) defined several $\mathcal{R}^-$ and several $\mathcal{R}^+$ methods which

compete to modify the solution in their *Adaptive Large Neighborhood Search* (ALNS) framework and applied their framework to the PDPTW. They implemented seven different strategies for selecting customers for removal: random, worst, related, cluster, time oriented, historical node-pair and historical request-pair. The solution is recreated by employing either greedy or regret insertion (Potvin and Rousseau 1993) with or without noise function. Pisinger and R pke (2007) applied ALNS to the Rich Pickup and Delivery Problem with Time Windows (RPDPTW).

Recent heuristic development based on the R&R principle builds upon the general ALNS framework by enlarging the set of $\mathcal{R}^-$ and $\mathcal{R}^+$ methods. Examples of such studies include, but are not limited to the papers by Laporte, Musmanno, and Vucelja (2010) and Ribeiro and Laporte (2012). In distinct contrast to the recent trend of introducing more and more $\mathcal{R}^-$ and $\mathcal{R}^+$ methods, the present paper introduces a streamlined R&R implementation: *Slack Induction by String Removals* (SISRs, phonetically “scissors”). This proposed solution approach employs a single $\mathcal{R}^-$ and a single $\mathcal{R}^+$ method: *adjacent string removal* and *greedy insertion with blinks*, respectively, with the significant additional benefit of being highly reproducible. An overview of these R&R methodological implementations is provided in Table 2.

Table 2 Overview of R&R methodologies.

<i>Shaw (1998) (LNS)</i>	
$\mathcal{R}^- \rightarrow$ related removal	$\mathcal{R}^+ \rightarrow$ optimal insertion (branch-and-bound)
<i>Schrimpf et al. (2000)</i>	
$\mathcal{R}^- \rightarrow$ random, radial or string removal	$\mathcal{R}^+ \rightarrow$ greedy insertion
<i>Pisinger and R�pke (2007) (ALNS)</i>	
$\mathcal{R}^- \rightarrow$ random, worst, related, cluster, time-oriented or historical removal	$\mathcal{R}^+ \rightarrow$ greedy or regret insertion with or without noise function
<i>Christiaens and Vanden Berghe (2018) (SISRs)</i>	
$\mathcal{R}^- \rightarrow$ adjacent string removal	$\mathcal{R}^+ \rightarrow$ greedy insertion with blinks

4. Local Search Metaheuristic

The SISRs neighborhood search, detailed in Section 5, is guided by *Simulated Annealing* (SA). This well known metaheuristic - introduced by Kirkpatrick, Gelatt, and Vecchi (1983) - is based on the statistical model concerning energy changes in annealing systems (Metropolis et al. 1953). In the present study, the system’s initial temperature $\mathcal{T}_0$ is reduced by the exponential cooling schedule

(Equation 1) to its final temperature $\mathcal{T}_f$ in f iterations. Cooling constant c is obtained by way of Equation 2.

$$\mathcal{T}_{k+1} = c \mathcal{T}_k \quad 0 \leq k < f \quad (1)$$

$$\text{where:} \quad c = \left(\frac{\mathcal{T}_f}{\mathcal{T}_0} \right)^{1/f} \quad \mathcal{T}_0 > \mathcal{T}_f > 0 \quad (2)$$

The probability h for accepting a neighbor solution s^* is dependent upon the current temperature $\mathcal{T}$ and the difference in energy $\Delta E = \text{dist}(s^*) - \text{dist}(s)$ relative to the current solution s (Equation 3). Neighbor solution s^* is accepted if a random selected value from the continuous uniform distribution $U(0,1)$ is less than the probability $h(\Delta E, \mathcal{T})$. Therefore, the acceptance criterion is expressed by way of Equation 4.

$$h(\Delta E, \mathcal{T}) = \exp(-\Delta E / \mathcal{T}) \quad (3)$$

$$\text{dist}(s^*) < \text{dist}(s) - \mathcal{T} \ln(U(0,1)) \quad (4)$$

A solution s is represented by two sets $s = \{T, A\}$. Set T is the set of tours, each of which serves at least one customer, while set A is the set of absent customers, all of whom are not served by any tour $t \in T$. The *initial solution* contains an equal number of tours as there are customers defined in the problem, therefore only one customer is served in each tour. This initial solution is passed to the local search procedure, detailed by way of Algorithm 1. First, the initial solution is saved as the best solution s^{best} found thus far (line 2). Temperature $\mathcal{T}$ is set to the initial temperature $\mathcal{T}_0$ (line 3). Next, f improvement iterations are performed (lines 4 to 10). During each iteration, a neighbor solution s^* is generated from current solution s by employing SISRs' R&R procedure (line 5). This neighbor is accepted as the new current solution s if the SA's acceptance criterion is satisfied (lines 6 and 7) and saved if it improves upon the best solution (lines 8 and 9). Finally the system is cooled down (line 10) before proceeding onto the next iteration.

Algorithm 1 Local search metaheuristic

Input: Initial solution $s = \{T, A\}$ where T is the set of tours and A the set of absent customers

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1: procedure LOCAL-SEARCH( $s$ )
2:    $s^{best} \leftarrow s$ 
3:    $\mathcal{T} \leftarrow \mathcal{T}_0$ 
4:   for  $f$  iterations do
5:      $s^* \leftarrow \text{SISRS-RUIN-RECREATE}(s)$ 
6:     if  $\text{dist}(s^*) < \text{dist}(s) - \mathcal{T} \ln(U(0, 1))$  then
7:        $s \leftarrow s^*$ 
8:     if  $\text{dist}(s^*) < \text{dist}(s^{best})$  then
9:        $s^{best} \leftarrow s^*$ 
10:     $\mathcal{T} \leftarrow c\mathcal{T}$ 
11: end procedure

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5. Slack Induction by String Removals (SISRs)

A brief explanation of SA and its employment for guiding SISRs is provided within Section 4. This section provides a detailed description of the SISRs method. The concepts of capacity slack, spatial slack and three premises are introduced in Section 5.1 which lay out the rationale for the proposed method. A detailed description of the SISRs' $\mathcal{R}^-$ and $\mathcal{R}^+$ method may be found in Section 5.2 and Section 5.3 respectively.

5.1. Rationale for SISRs

When customers are removed from a tour's sequence, several types of slack may be introduced. First, the most apparent type of slack may be *capacity slack* referring to the vehicle's capacity which is set free. Second, *spatial slack* may be identified if one imagines that the vehicle's travel distance is limited. Given this limited travel distance, the vehicle's reachable area may be defined. This area appears as an elliptical region around each edge, where each edge's end points are the focus points for its corresponding ellipse. This area increases when the locations of the removed customers are no longer visited and we refer to this increase as spatial slack. While the reduced travel distance contributes the most for gaining spatial slack, it is not a necessary condition. For example, removing the intermediate of three collinear locations does not reduce the travel distance while the reachable area - spatial slack - does increase given the larger focal distance. Although, the vehicle's travel distance is not limited within the CVRP definition, one may express

spatial slack relative to the vehicle's original tour, as the additional reachable area delimited by its original travel distance.

An effective $\mathcal{R}^-$ method should introduce both types of slack into the solution such that the possibilities for improving the solution are greatly enhanced. The following premises serve as the basis for such $\mathcal{R}^-$ method. Each premise addresses the potential failures of alternative $\mathcal{R}^-$ methods, within the context of slack induction.

PREMISE 1. Remove a ‘sufficient’ number of customers

Removing a small number of customers (such as in Fig. 2(a₁)), may fail to introduce sufficient capacity slack in the vehicles for serving customers of higher demand. While it is visually clear (a₂) by which vehicles the removed customers should be served, it may be impossible if the vehicles gained insufficient capacity slack.

PREMISE 2. Remove ‘adjacent’ customers

Removing customers which are not geographically close (adjacent) to each other (such as randomly-selected customers in Fig. 2(b₁)), generally introduces capacity and spatial slack scattered across many tours (b₂). Such slack is unlikely to be beneficial when the solution is recreated. The recreated solution may be comparable to the result of multiple ruin and recreate iterations wherein a small number of customers is removed each time.

PREMISE 3. Remove ‘strings’ of customers

Removing customers of which their selection is only based on adjacency (as in the case of radial removal, Fig. 2(c₁)) may often fail to capture complete detours which remain present in the ruined state (c₂). This may be remedied by increasing the radius, however, much more customers than strictly necessarily may be removed as well. Therefore, removing strings may be much more effective.

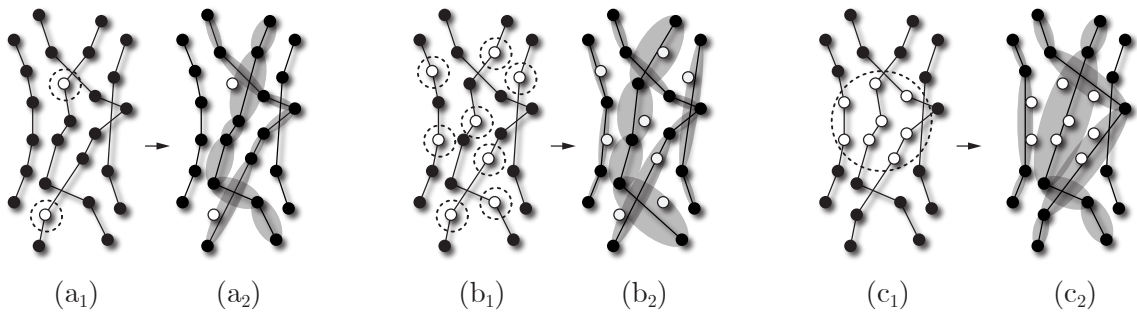


Figure 1 Examples of ruin methods which introduce insufficient slack: removal of insufficient customers (a), randomly-selected customers (b) and radially-selected customers (c). Figures indexed by 1 illustrate the partial solution wherein the customers selected for removal are colored white and outlined by dashed lines. Figures indexed by 2 illustrate the ruined state wherein spatial slack relative to the original state is shaded gray..

If one would remove multiple strings which are geographically close to each other - removal of adjacent strings - a very effective yet simple of origin $\mathcal{R}^-$ method may be obtained. An example of such adjacent string removals is depicted in Fig. 2. Only two adjacent strings are selected for removal in (a) and the ruined state of the solution is illustrated in (b). Notice that (i) spatial slack is greatly introduced in only two tours and (ii) the spatial slack of both tours greatly overlap with one another. Ideally, the ruined state obtained by the $\mathcal{R}^-$ method (b), is recreated by the $\mathcal{R}^+$ method into the solution illustrated in (c).

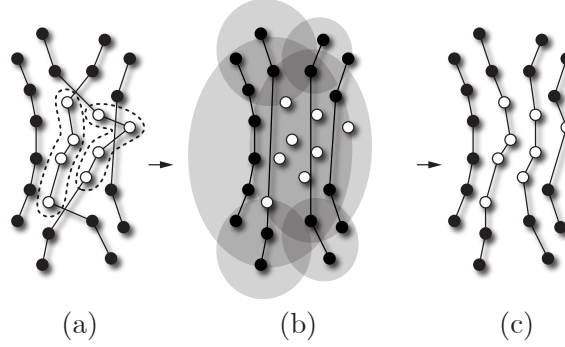


Figure 2 An example of adjacent string removals: selections of two adjacent strings (a), ruined stated with great spatial slack induced (b) and an improved recreated state (c).

5.2. SISRs' ruin method - Adjacent string removal

The proposed $\mathcal{R}^-$ method is based on Premises 1 to 3 as follows. The number of removed customers is controlled by means of the average number of removed customers $\bar{c}$. In addition, the $\mathcal{R}^-$ method may be set for removing many strings of small cardinality or few strings of high cardinality by means of the maximum cardinality of the removed strings L^{max} . Let s be the solution being ruined. A valid number of removed strings k_s and cardinality l_t for every string (each of which are removed from a different tour t) is selected as follows. The maximum string cardinality l_s^{max} is initially set to the minimum of L^{max} and the solution's average tour cardinality $\overline{|t \in T|}$ (Eq. 5) before the maximum number of strings k_s^{max} is derived (Eq. 6).

$$l_s^{max} = \min\{L^{max}, \overline{|t \in T|}\} \quad l_s^{max} \in \mathbb{R}^+ \quad (5)$$

$$k_s^{max} = \frac{4\bar{c}}{1 + l_s^{max}} - 1 \quad k_s^{max} \in \mathbb{R}^+ \quad (6)$$

$$k_s = \lfloor U(1, k_s^{max} + 1) \rfloor \quad k_s \in \mathbb{N}^+ \quad (7)$$

The number of strings k_s to be removed is obtained by rounding down a randomly selected real value from the continuous uniform distribution $U(K^{min}, k_s^{max} + 1)$ by way of Eq. 7. The maximum cardinality of each string is limited by the initial limit l_s^{max} and the tour's cardinality $|t|$ from which

the string is to be removed (Eq. 8). Finally, the cardinality for the string to be removed from tour t is obtained by rounding down a real value selected from $U(L^{min}, l_t^{max} + 1)$ Eq. 9.

$$l_t^{max} = \min\{|t|, l_s^{max}\} \quad l_t^{max} \in \mathbb{R}^+ \quad (8)$$

$$l_t = \lfloor U(1, l_t^{max} + 1) \rfloor \quad l_t \in \mathbb{N}^+ \quad (9)$$

Before the procedural steps of the ruin phase $\mathcal{R}^-$ are detailed by way of Algorithm 2, two assumptions are made: (i) For each customer c an adjacency list $adj(c)$ is assumed to be available for identifying adjacent strings. The list contains all customers ordered by increasing distance from c , which includes c as its first element. (ii) Each string being removed originates from a different tour such that spatial slack as well as capacity slack is spread across a number of tours, controlled by k_s . Therefore a tour may be ruined at most once and removing k_s strings implies ruining k_s tours.

Algorithm 2 SISRs' ruin

Input: Solution $s = \{T, A\}$ where T is the set of tours and A the set of absent customers

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1: procedure RUIN( $s$ )
2:    $l_s^{max}, k_s^{max}, k_s \leftarrow \text{calculate}(\text{Eq. 5, 6 and 7})$ 
3:    $c_s^{seed} \leftarrow \text{randomCustomer}(s)$ 
4:    $R \leftarrow \emptyset$ 
5:   for  $c \in adj(c_s^{seed})$  and  $|R| < k_s$  do
6:     if  $c \notin A$  and  $t \notin R$  then
7:        $c_t^* \leftarrow c$ 
8:        $l_t^{max}, l_t \leftarrow \text{calculate}(\text{Eq. 8 and 9})$ 
9:        $A \leftarrow A \cup \text{removeSelected}(t, l_t, c_t^*)$ 
10:       $R \leftarrow R \cup t$ 
11: end procedure

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The procedural steps of the ruin phase $\mathcal{R}^-$ are given by Algorithm 2. First, values for l_s^{max} , k_s^{max} and k are derived by Equations 1 to 3 (line 2). A randomly selected customer serves as the seed customer c_s^{seed} from which string removals are initiated (line 3). The auxiliary set R , for storing ruined tours, is initially empty (line 4). The adjacency list from c_s^{seed} is iterated over from the first customer to the last until set R contains k_s ruined tours indicating that k_s strings have been removed (lines 5 to 10). In each iteration, if the customer c has not been removed ($c \notin A$)

and its serving tour t has not been ruined ($t \notin R$) (line 6), a string is removed from this tour (lines 7 to 10). Such a customer is special in the sense that it is the closest customer to c_s^{seed} out of all customers served by t , denoted as c_t^* (line 7). The maximum cardinality l_t^{max} and final cardinality l_t of the string to be removed from tour t is obtained by way of Equations 4 and 5 (line 8). The string is removed via the ‘string’ or ‘split string’ procedure based on c_t^* and l_t . In either case, l_t customers are selected and removed from t which are added to the set A (line 9) and the ruined tour is added to set R (line 10).

The ‘string’ procedure removes a randomly selected string of cardinality l_t in tour t which includes customer c_t^* . By removing a string which includes c_t^* , other customers inside this string are very likely to be located closely to c_{seed} as well. Therefore, every call to this procedure will result in a string removal of customers close to the seed customer or simply, strings adjacent to each other. An example where $l_t = 3$ is illustrated in Fig. 3. One of the three possible strings is randomly selected for removal.

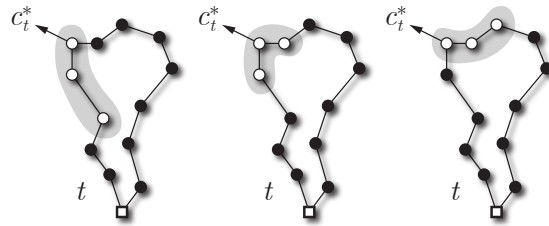


Figure 3 Possible ‘string’ removals which include c_t^* from tour t when $l_t = 3$.

The ‘split string’ procedure begins much like ‘string’, by randomly selecting a string of cardinality $l + m$ which includes customer c_t^* (Fig. 4a). However, the ruin phase bypasses and preserves a random substring of m intervening customers as shown in Fig. 4b. The number of preserved customers m is determined as follows. Initially $m = 1$ and the current value of m is maintained if a random number is smaller than α ($U(0,1)$) or when the maximum value for m is reached ($m = m^{max} = |t| - l$). If neither of these conditions is satisfied, m is incremented ($m = m + 1$) and the incrementation process repeats.

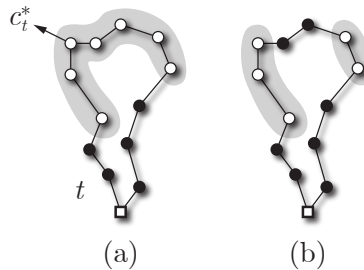


Figure 4 ‘Split string’ removal when $l = 5$ and $m = 2$.

Both ‘string’ and ‘split string’ are executed with equal probabilities of which an example is illustrated in Fig. 5a for ‘string’ and (b₁), (b₂) for ‘split string’ with small and large m -values. Customers being removed are colored white. In half of cases the ‘string’ procedure removes a string, introducing spatial slack (Fig. 5a). In the other half, tours are ruined by the ‘split string’ procedure. In this ‘split string’ removal procedure, a small value for m is obtained with a very low probability. When this occurs, customers close to c_t^* are removed while remaining spatially bonded to the same regions as the preserved customers (b₁). Otherwise $m = m^{max}$ is obtained with much greater probability, indicating that the customers removed are close to the depot (b₂).

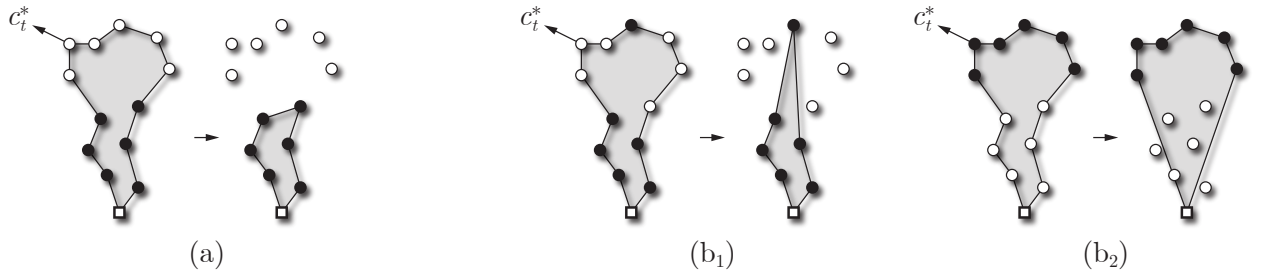


Figure 5 ‘String’ removal (a) and ‘split string’ removal when m equals the lowest (b₁) or highest (b₂) possible value.

Fig. 6 presents an example of the ruin phase where two strings ($k_s = 2$) are removed from solution s . First, the seed customer c_s^{seed} is randomly selected (a). Following this, the list $adj(c_s^{seed})$, is iterated over. Since c_s^{seed} is always the first element in the list and no tours are ruined, $c_{t1}^* = c_s^{seed}$. A string of length $l_{t1} = 4$ is removed by the ‘string’ procedure which includes c_{t1}^* (b-c). In (d) $adj(c_s^{seed})$ is iterated over until the next customer (c_{t2}^*) is found in a not-yet-ruined tour. The second string is also removed by the ‘string’ method and l_{t2} is, coincidentally, 4 again (e). The final ruined state is illustrated in (f).

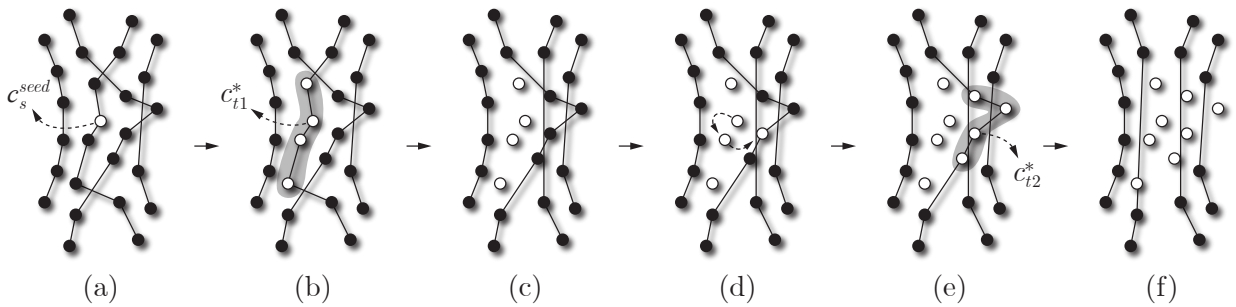


Figure 6 An example of the SISRs ruin method.

5.3. SISRs' recreate method - Greedy insertion with blinks

While basic greedy inserts each customer sequentially at the best position, one may deviate slightly from the best position. Pisinger and R pke (2007) apply *greedy insertion with noise function* by adding a randomized noise term to the insertion cost. This paper, by contrast, introduces *greedy insertion with blinks* (Algorithm 3).

First, the set of absent customers A is sorted by one of the following orders: *Random*, *Demand*, *Far* or *Close* (line 2). ‘Random’ enables the set’s insertion without any ordering. ‘Demand’ sorts customers by demand, placing those with the largest demand first. ‘Far’ inserts the most-distant customers from the depot first. Finally, ‘Close’ inserts customers closest to the depot first. Set A is sorted by Random, Demand, Far and Close by weights equal to four, four, two and one, respectively. Each customer $c \in A$ (line 3) is inserted into solution s at the ‘best’ position P as follows. Initially, P is set to NULL, indicating that no position for c is selected (line 4). All current tours which are part of the solution ($t \in T$) are iterated over in a random order (line 5). When a tour t has sufficient capacity slack to serve c , all positions P_t inside this tour are iterated over (line 6). Each position is evaluated with a probability of $1 - \beta$, otherwise skipping the position as if the algorithm ‘blinks’ and therefore β represents the blink rate (line 7). If a position P_t is found for which the cost of inserting c is lower than the current best position P , P_t becomes the new best position (lines 8 and 9). If no position was found in existing tours, a new empty tour is created to serve c (lines 10 to 12). Finally c is inserted at P and removed from the the set of absent customers (lines 13 and 14). Blink rate $\beta = 0$ denotes that customers are always inserted at the best position, a strategy proven to be sub-optimal after experimentation. This may appear somewhat counter-intuitive given that it is more logical to assume that consistently inserting each customer at the best possible position would result in the highest quality solution.

Algorithm 3 SISRs' recreate

Input: Ruined solution $s = \{T, A\}$ where T is the set of tours and A the set of absent customers

```

1: procedure RECREATE( $s$ )
2:    $sort(A)$ 
3:   for  $c \in A$  do
4:      $P \leftarrow \text{NULL}$ 
5:     for  $t \in T$  (which can serve  $c$ ) do
6:       for  $P_t$  in  $t$  do
7:         if  $U(0, 1) < 1 - \beta$  then
8:           if  $P = \text{NULL}$  or  $costAt(P_t) < costAt(P)$  then
9:              $P \leftarrow P_t$ 
10:    if  $P = \text{NULL}$  then
11:       $T \leftarrow T \cup \text{new tour } t$ 
12:       $P \leftarrow \text{position in } t$ 
13:    insert  $c$  at  $P$ 
14:     $A \leftarrow A \setminus \{c\}$ 
15: end procedure

```

Out of all feasible options for inserting a customer, one may make a random choice or utilize a heuristic to pick the best option (greedy insertion). The effect of blinks is reflected by the probability p of selecting a specific option based on its rank r . A blink rate of β implies the best option, which is ranked first ($r = 1$), is selected with probability $p(1) = (1 - \beta)$. If the best ranked option is blinked over, the probability of selecting the second best option is $p(2) = (1 - \beta)\beta$. Selection probability of the third ranked option equals $p(3) = (1 - \beta)\beta^2$. The selection probability for each rank is expressed by exponential function:

$$p(r) = (1 - \beta) \beta^{(r-1)} \quad r \in \{1, \dots, \infty\} \quad (10)$$

Notice how the blinking algorithm itself is unaware of each option's rank, it only blinks with probability β while iterating over all options. Thus, blinking results in rank-based selection probabilities without requiring one to rank the options first, in contrast to *heuristic-biased stochastic sampling* (HBSS) (Bresina 1996), a closely related selection algorithm which is compared against the blinking algorithm in Appendix A.

6. Fleet Minimisation

This work introduces a two-stage approach to accommodate a hierarchical objective which is primarily the minimisation of the number of vehicles as follows. The newly developed *absences-based* neighbor acceptance criterion is employed during the vehicle minimisation stage, detailed by way of Algorithm 4. The solution which serves all customers with the lowest number of vehicles is represented by s^{best} (line 2). For each customer c the number of solutions where c was not served by any tour (absent from the solution) is tracked by its associated absence counter abs_c . This counter is initially set to 0 for all customers (lines 3 and 4). In each iteration the following steps are performed. First, a neighbor solution s^* is obtained by the SISRs' $\mathcal{R}^-$ and $\mathcal{R}^+$ procedure (line 6). This neighbor is accepted if (i) the number of absent customers decreased or (ii) the sum of absence counters associated with the absent customers $sumAbs(A) = \sum_{c \in A} (abs_c)$ decreased (lines 7 and 8). If the current solution is feasible ($A^* = \emptyset$), a copy is saved as s^{best} . Afterwards the tour whose served customers represent the lowest sum of absence counters is removed from the current solution (lines 9 to 11). Finally, each absent customer in the neighbor solution has their associated absence counter incremented (lines 12 and 13).

Algorithm 4 Fleet size minimisation

Input: Solution $s = \{T, A\}$ where T is the set of tours and A the set of absent customers

```

1: procedure FLEET-MINIMISATION( $s$ )
2:    $s^{best} \leftarrow s$ 
3:   for  $c$  of the problem do
4:      $abs_c \leftarrow 0$ 
5:   while minimising tours do
6:      $s^* \leftarrow \text{SISRS-RUIN-RECREATE}(s)$ 
7:     if  $|A^*| < |A|$  or  $sumAbs(A^*) < sumAbs(A)$  then
8:        $s \leftarrow s^*$ 
9:     if  $A^* = \emptyset$  then
10:       $s^{best} \leftarrow s^*$ 
11:      remove  $t \in T$  with the lowest  $sumAbs(t)$ 
12:     for  $c \in A^*$  do
13:        $abs_c \leftarrow abs_c + 1$ 
14: end procedure

```

In essence, neighbor s^* is accepted only if the number of absent customers $|A^*|$ decreased or if the sum of absence counters associated with the absent customers $sumAbs(A^*)$ decreased. This *absences-based* acceptance criterion for minimising fleet size proved to be very efficient without complicating the overall method. In fact, it only requires the replacement of SA's *distance-based* acceptance criterion by this *absences-based* acceptance criterion during the fleet minimisation phase.

7. Parameters

During the development of the proposed algorithm, the following parameters were set. The parameters are all fixed and then one by one varied to determine their influence on the algorithm's performance. Table 3 presents the superior parameter settings employed throughout the experiments reported in Section 8.

Table 3 SISRs' and SA's parameter settings.**SISRs's $\mathcal{R}^-$**

$$\bar{c} = 10$$

An average of 10 removed customers has been proven a good ruin parameter. When $\bar{c}$ is equal to 10 it implies the removal of at least one and at most 19 customers. Decreasing this parameter, for example to five, was observed to be insufficient for diversification. By contrast, increasing $\bar{c}$ for example to 20, results in extensively ruined solutions. Such solutions cannot easily be recreated into improving solutions and most likely result in highly divergent neighbors, which disables intensification of potentially good solutions.

$$L^{max} = 10$$

The cardinality of removed strings is limited to 10 since removing longer strings fails to improve final solution quality. This confirms the observations made by Fahrion and Wrede (1990) whose chain-exchange procedure limits string cardinality to half the average tour length. By employing Equation 7 the number of removed strings k_s will range from one to three for solutions with large tours ($|t \in T| \geq L^{max}$) and one to 19 for solutions whose tours all have a cardinality of one ($|t \in T| = 1$). This situation almost always only occurs during the first iteration of the search (see Section 4).

$$\alpha = 0.01$$

This setting results in values for the number of preserved customers $m = 1, 2, 3, \dots, m^{max}$ having respective probabilities $p = 1.00\%, 0.99\%, 0.9801\%, \dots, (100 - p_{m=1} - p_{m=2} - p_{m=3} - \dots - p_{m=m^{max}-1})\%$. As such, it is very likely that customers removed within a 'split string removal' are those at the start or end in the tour's sequence.

SISRs's $\mathcal{R}^+$

$$\beta = 0.01$$

It was observed that a blink rate of 1% ($\beta = 0.01$) was the most effective in terms of final solution quality. A result that may, at first, appear somewhat counter-intuitive given that it is more logical to assume that consistently inserting each customer at the best possible position would result in the highest quality solution.

Fleet minimisation

$$iterations = 10\% \text{ of all iterations}$$

Generally the minimum number of vehicles is found within three minutes. Therefore, dedicating 10% of all iterations to the fleet minimisation is considered to be sufficient.

SA

$$\mathcal{T}_0 = 100, \mathcal{T}_f = 1$$

The search begins at an initial temperature $\mathcal{T}_0 = 100$ and ends at the final temperature $\mathcal{T}_f = 1$.

$$iterations = it(v)$$

The number of iterations it is determined as a function of problem size v by linear interpolation. The minimum and maximum problem size is 100 and 1000 respectively, as per Uchoa et al. (2017). The present study set $it(100) = 3 \times 10^7$ and $it(1000) = 3 \times 10^8$, thus enabling the direct comparison of SISRs' calculation times against those from the aforementioned paper.

Performance of Algorithm 4 is tested on the 1000 customers benchmark set of Gehring and Homberger (2001) by taking the best of five runs per instance. The cumulative number of vehicles (CNV) for each set is compared against the (best of five) CNV obtained by NB-f(n) (Nagata,

Bräysy, and Dullaert 2010) and HGSADC (Vidal et al. 2013) in Table 4. In addition, results of SISRs with different L^{max} settings are tested. The best results are marked dark gray while results of SISRs are marked light gray when better than the competing algorithms but not the best of the different L^{max} settings. The average calculation time (T) is specified in minutes which is at most 10 minutes for NB-f(n) and unknown for HGSADC.

Table 4 Fleet minimisation. Results are ranked first (best), second and third place by use of the colors dark gray, gray, and white respectively.

		NB-f(n)	HGSADC	SISRs		
				$L^{max} = 10$	$L^{max} = 5$	$L^{max} = 2$
Set	n	CNV (T)	CNV	CNV (T)	CNV (T)	CNV (T)
R1	1000	919 (≤ 10)	919	919 (0.2)	919 (0.2)	919 (0.1)
R2	1000	190 (≤ 10)	190	190 (0.0)	190 (0.0)	190 (0.0)
C1	1000	941 (≤ 10)	941	939 (2.8)	938 (2.4)	939 (2.7)
C2	1000	291 (≤ 10)	288	291 (9.1)	292 (3.7)	288 (2.5)
RC1	1000	900 (≤ 10)	900	900 (0.1)	900 (0.0)	900 (0.0)
RC2	1000	183 (≤ 10)	182	182 (0.3)	182 (0.3)	188 (0.4)

These results indicate that different settings may be beneficial for certain sets. For example, removing more strings of lower cardinality ($L^{max} = 2$) enables to find the best CNV for set C2 while removing strings of intermediate cardinality ($L^{max} = 5$) obtained a better CNV than NB-f(n) and HGSADC for set C1. Although, this parameter tuning may improve the results for these problems, the same parameter setting (presented in Table 3) employed for solving the CVRP and each of its variants addressed in Section 8.1 and 8.2 is used for solving the VRPTW and PDPTW.

8. Computational Results

SISRs' computational results are documented throughout the following three subsections. First, SISRs' results for the most relevant CVRP benchmark set are compared against those obtained by current state-of-the-art methods within Section 8.1. Following this, although the CVRP constitutes the present work's core focus, SISRs' general applicability is verified by way of solving numerous CVRP variants. The very minor adjustments necessary to enable SISRs to solve each of these problems is detailed within Section 8.2.

8.1. CVRP

Uchoa et al. (2017) introduced a new benchmark set due to the fact that *“the existing sets became too easy, are too artificial or do not cover the wide range of characteristics found in real applications”*. This recent benchmark set contains 100 instances whose sizes range from 100 to 1000

customers and which cover a wide variety of characteristics. Uchoa et al. (2017) report the detailed results obtained by two state-of-the-art heuristics, both conducted on Xeon CPU @ 3.07 GHz:

- ILS-SP: Iterated Local Search with Set Partitioning of Subramanian, Uchoa, and Ochi (2013)
- UHGS: Unified Hybrid Genetic Search of Vidal et al. (2014)

This study's experiments were performed on Xeon(R) CPU E5-2650 v2 @ 2.60GHz, and compared against the results of ILS-SP and UHGS which were taken from Uchoa et al. (2017). No reimplementation of these methods was therefore required. The full benchmark set is divided into three sets for comparison based on problem size n : Small, Medium and Large (Table 5).

Table 5 Complete benchmark subsets.

	n	Instances	# Instances
Small	100 – 250	X-n101-k25 – X-n247-k47	32
Medium	250 – 500	X-n251-k28 – X-n491-k59	36
Large	500 – 1000	X-n502-k39 – X-n1001-k43	32
All	100 – 1000	X-n101-k25 – X-n1001-k43	100

The complete results obtained by each heuristic are documented in Appendix B, where Tables 11, 12 and 13 report the results for the small, medium and large instances, respectively. The average distance (Avg), the best distance (Best), the average calculation time in minutes (T) and the average gap to the best known solution in percentage terms (Gap) on 50 runs per instance are presented in these tables. Heuristics are ranked first (best), second and third place by use of the colors dark gray, gray, and light gray respectively for the columns Avg, Best and T. Additionally, the distance (Dist), number of vehicles (NV) and optimality (Opt) (proven by Pecin et al. (2014)) of the best known solution (BKS) are also reported. Throughout the entire experimental campaign, 40 BKSs were improved and 40 solutions of equal quality were found which are marked dark and light gray respectively in the BKS column.

Results presented in each column (Avg, Best and T) of Tables 11, 12 and 13 have been summarized in the corresponding columns (Avg, Best and T) of Table 6 by reporting the minimum (Min), maximum (Max), average (Avg) and median (Median) value. Results in columns Avg and Best of Table 11, 12 and 13 are expressed as gaps in percentages relative to the best known solutions reported by Uchoa et al. (2017) before summarizing them in Table 6.

Table 6 Summarized computational results. Results are ranked first (best), second and third place by use of the colors dark gray, gray, and light gray respectively.

	ILS-SP			UHGS			SISRs		
	Avg	Best	T	Avg	Best	T	Avg	Best	T
Small (100-250)									
Min	0.00	0.00	0.1	0.00	0.00	1.4	0.00	0.00	0.2
Max	2.50	1.19	17.8	0.30	0.06	20.4	0.50	0.16	18.4
Avg	0.31	0.12	2.4	0.07	0.00	6.0	0.11	0.01	5.2
Median	0.19	0.00	1.6	0.03	0.00	5.4	0.05	0.00	5.0
Medium (250-500)									
Min	0.00	0.00	2.0	0.00	0.00	6.5	0.01	-0.18	9.8
Max	1.54	0.73	60.6	0.49	0.10	86.7	1.35	0.28	58.4
Avg	0.54	0.17	23.1	0.24	0.02	30.3	0.20	0.02	27.0
Median	0.42	0.12	16.8	0.26	0.00	22.4	0.15	0.01	22.4
Large (500-1000)									
Min	0.00	0.00	27.3	0.11	0.00	33.1	-0.26	-0.42	60.9
Max	1.74	1.45	792.8	0.68	0.33	560.8	0.38	0.15	412.7
Avg	0.76	0.49	195.7	0.29	0.04	268.6	-0.01	-0.13	152.0
Median	0.75	0.39	144.7	0.26	0.02	258.6	0.01	-0.14	144.8
All (100-1000)									
Min	0.00	0.00	0.1	0.00	0.00	1.4	-0.26	-0.42	0.2
Max	2.50	1.45	792.8	0.68	0.33	560.8	1.35	0.28	412.7
Avg	0.53	0.26	71.7	0.20	0.02	98.8	0.10	-0.03	60.0
Median	0.40	0.11	17.7	0.20	0.00	22.4	0.07	0.00	22.4

UHGS outperforms the other methods for Small instances. For the Medium set, UHGS and SISRs perform equally well insofar as obtaining the best solutions (Best). Additionally, it was observed that SISRs was the most robust method. Meanwhile, for the Large set, SISRs outperforms both heuristics on all aspects - it is the most robust method which finds the best solutions in the least amount of computation time.

The average solution quality (Avg columns of Table 11, 12 and 13) expressed as gaps have also been visualized by way of boxplots for each heuristic and problem set in Fig. 7. These boxplots demonstrate SISRs' efficiency and robustness by exhibiting small median and spread of average gaps.

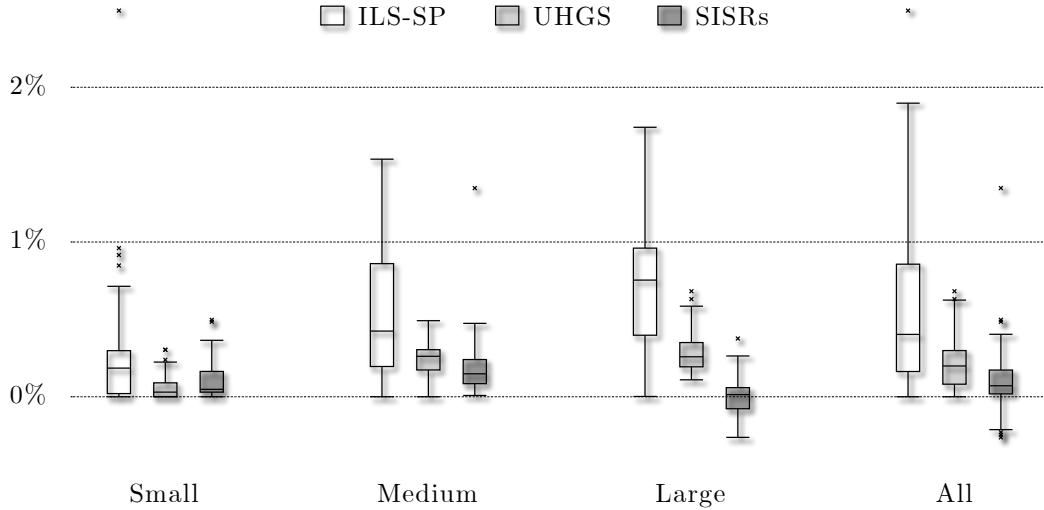


Figure 7 Gaps of the average results to the best known solutions (Uchoa et al. 2017) visualized by way of boxplots.

Finally, average solution quality is analyzed by conducting a one-tailed Wilcoxon signed-rank test (Wilcoxon 1945), which is a nonparametric test for comparing two independent groups and performed by the software environment R (R Core Team 2017). The only assumption typically required for this test is that observations are independent. The following null-hypothesis H_0 and alternative hypothesis H_1 is considered, where X represents ILS-SP or UHGS.

$$H_0 : \text{Avg}(\text{SISRs}) = \text{Avg}(X)$$

$$H_1 : \text{Avg}(\text{SISRs}) > \text{Avg}(X)$$

The average results obtained by SISRs are compared against the average results of ILS-SP and UHGS. Rejecting H_0 implies that the average results obtained by SISRs are different from those obtained by alternative heuristic X . Rejecting H_1 implies that the values of the average results obtained by SISRs are lower than those obtained by X or that SISRs' average solution quality is better than X '. The obtained p-values are presented in Table 7.

Table 7 P-values of the Wilcoxon rank-sum test.

	Small		Medium		Large		All	
	ILS-SP	UHGS	ILS-SP	UHGS	ILS-SP	UHGS	ILS-SP	UHGS
H_0	0.029239	0.022526	0.000013	0.002624	0.000000	0.000000	0.000000	0.000000
H_1	0.014620	0.989361	0.000007	0.001312	0.000000	0.000000	0.000000	0.000000

While the overall significance level α is set to 2.5%, Bonferroni correction (Dunn 1961) is applied in order to compensate for the family-wise error rate. Bonferroni correction sets the significance

level for testing individual hypothesis to α/m where m is the number of tests performed using the same data. Since four tests are performed using the Small, Medium and Large data sets and four again using All data, a total of eight hypotheses are tested each time with the same data. Therefore p-values are assessed with respect to the Bonferroni adjusted level $0.025/8 = 0.003125$ or 0.3125%. If the p-value is lower than this level, the hypothesis is rejected and the cell is colored gray. The null-hypothesis H_0 and alternative hypothesis H_1 are both rejected when considering Medium, Large and All benchmark sets indicating that SISRs significantly differs from both other heuristics and that its average results are significantly better.

8.2. CVRP Variants

Minimal adjustments were necessary to enable SISRs to solve the CVRP variants addressed and therefore a structured overview for the adjustments related to each problem attribute is provided by way of Table 8. If more than one change was necessary then the changes take the form of a numbered list.

Various sets of benchmark instances from the literature were employed to evaluate SISRs' performance. Its results were then compared against state-of-the-art methods for each CVRP variant. A detailed breakdown of the results obtained by both SISRs and these state-of-the-art methods for each individual instance is presented in Appendix C. A causal consultation of these results immediately reveals SISRs' impressive dominance. Despite the various state-of-the-art methods it is compared against, being general frameworks or dedicated single-problem heuristics, SISRs' proves highly competitive: very often matching or improving the best known solutions across all variants. Even more impressively, for large-scale instances - which are more representative of real-world environments - SISRs produces many new best known solutions. This is especially true for the PDPTW, for which SISRs improves more than one third of the considered benchmark set's instances.

Table 8 Adjustments accommodating VRP variants.

Multi-Depot
1) ($\mathcal{R}^+$) While sorting customers by ‘Far’ or ‘Close’ to depot (see Section 5.3), the distance to the closest depot is employed. 2) ($\mathcal{R}^+$) When a new vehicle must be used to include customer c , the closest depot to c which has an unused vehicle is selected.
Backhauls
1) ($\mathcal{R}^-$) If a tour only contains backhaul customers after a string is removed, all remaining customers are removed. 2) (SA) A solution associated with more tours which exclusively contain backhaul customers is never accepted.
Mixed/Simultaneous Deliveries and Pickups.
In each tour stop two additional values are maintained which correspond to the maximum capacity consumption in all previous/following stops. This enables a capacity check in constant time ($\mathcal{R}^+$).
Open
A dummy location to which all other locations have a distance equal to zero is introduced. This location represents the end depot in each tour.
Cumulative
($\mathcal{R}^+$) The delta evaluation method, which is employed to select the best insertion position, is modified to represent the cumulative objective.
Service Time
($\mathcal{R}^+$) An additional ordering in terms of decreasing service time is employed when each customer has an individual service time.
Fleet Minimisation
1) Minimisation of fleet size is performed by employing the <i>absences-based</i> acceptance criterion during the fleet minimisation phase which is executed during the first 10% of iterations. Algorithm provides details concerning this procedure. 4. 2) A solution with an increased number of tours is never accepted in the distance minimisation phase.
Time Windows
1) Feasibility checks concerning time windows are based on the ‘push forward’ global variables (Savelsbergh 1990). 2) ($\mathcal{R}^+$) In addition to the sorting orders introduced in Section 5.3 (Random, Demand, Far and Close), customers are sorted with respect to increasing time window width, increasing time window start and decreasing time window end.
Pickup & Deliveries
1) ($\mathcal{R}^-$) Strings are removed as described in Section 5.2. In addition, the associated pickup/delivery nodes which were not included by the string are also removed. 2) ($\mathcal{R}^+$) Greedy insertion with blinks (Algorithm 3) is modified to iterate over feasible combinations for pickup and delivery insertion positions.

9. Conclusions

This paper introduced SISRs, a heuristic procedure capable of achieving highly competitive and new best results within short computation times for both the CVRP and a wide range of associated problems such as the VRP with Time Windows and the Pickup & Delivery problem. SISRs is composed of a ruin method which generates spatial slack, followed by a greedy recreate heuristic

which leaves sufficient room for solution space exploration. In addition to these components, this paper also contributes a fleet minimization approach to VRP variants focusing primarily on minimising their number of vehicles employed.

In many respects Laporte (2009) provided much of the motivation behind this paper’s original contribution when writing of how *“There is ... a sense that several of the most successful metaheuristics are over-engineered and one should now attempt to produce simple and flexible algorithms capable of handling a larger variety of constraints, even if this were to translate into a small loss in accuracy”*. While SISRs exhibits the simplicity and flexibility Laporte recommends, it also serves as a counter to argument that this necessitates a loss in accuracy. As demonstrated by this paper, no such trade-off occurs. Through minimal algorithmic modification, SISRs obtains highly competitive and new-best solutions for the wide variety of problems addressed, consistently competitive with highly-specialized methods developed specifically for individual problems.

Both SISRs and the fleet minimization approach indeed achieve a previously challenging reproducibility while simultaneously demonstrating their ability to generate high quality solutions for a wide range of VRP variants. Furthermore, performance with respect to large-scale academic instances was particularly impressive and exceeded expectations. Therefore, there is also room for further research in terms of even larger instances that are more representative of real-world conditions, the results for which should be broadly similar to those obtained for the largest instances addressed within the present study. Finally, all the newly introduced heuristic components demonstrate an impressive convergence speed which engenders an array of future real-world applications, providing a multitude of exciting avenues for future research.

Acknowledgments

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Appendix A: Blinks and HBSS

Iterative sampling (Langley 1992) builds an empty solution into a complete solution, which represents a ‘sample’ of the solution space, by incrementally making random decisions at each decision point. For example when applied to VRP, an unserved customer is randomly selected and inserted at a random position in the partial solution until all customers have been inserted. By contrast, decisions may be made based solely on a specific heuristic. For example, an unserved customer is inserted at the best position by *greedy insertion*. Meanwhile, in *Heuristic-biased Stochastic Sampling* (HBSS) (Bresina 1996), the heuristic’s preferred options are applied with a certain probability as follows. The heuristic ranks all R options at a decision point. A rank-based weight $bias(r)$ is assigned to each option by way of a specific *bias function*. The probability $p(r)$ of selecting an option based on its rank r is obtained by dividing its weight by the sum of all option weights:

$$p(r) = \left(\sum_{k=1}^R bias(k) \right)^{-1} bias(r) \quad r \in \{1, \dots, R\} \quad (11)$$

Table 9 presents the rank-based probabilities obtained by the following bias functions: constant, logarithmic, linear, cubic and exponential. For each bias function, the left column represents $p(r)$ assuming there exist only five options ($R = 5$) at the considered decision point. Thirty options are assumed ($R = 30$) within the right hand column and the remaining probability of selecting one of the options $r = 6..30$ is detailed in the last row.

Table 9 HBSS rank selection probabilities.

	Constant		Logarithmic		Linear		Cubic		Exponential	
r	1		$\log^{-1}(r+1)$		r^{-1}		r^{-3}		e^{-r}	
1	0.200	0.033	0.339	0.109	0.438	0.250	0.843	0.832	0.636	0.632
2	0.200	0.033	0.214	0.069	0.219	0.125	0.105	0.104	0.234	0.233
3	0.200	0.033	0.170	0.055	0.146	0.083	0.031	0.031	0.086	0.086
4	0.200	0.033	0.146	0.047	0.109	0.063	0.013	0.013	0.032	0.031
5	0.200	0.033	0.131	0.042	0.088	0.050	0.007	0.007	0.012	0.012
6-30	-	0.833	-	0.678	-	0.429	-	0.013	-	0.006

Clearly, probabilities are dependent upon the number of options. With regard to the logarithmic bias function, the probability $p(r)$ of selecting the best option $r = 1$ is three times smaller when thirty options are available (presented in the right column) compared to when only five options are available (presented left). This influence is negligible when the bias function quickly converges to zero and the partial sum $\sum_{k=1}^R bias(k)$ converges to the infinite sum ($R = \infty$). This occurs for strong bias functions, such as the cubic and exponential bias function.

In contrast to HBSS, *greedy insertion with blinks*, introduced by the present paper in Section 5.3, does not require one to explicitly rank the options in advance to obtain rank-based selection probabilities. Nevertheless,

the exponential rank-based probability function of the blinking algorithm (Eq. 10 and 12) may be expressed as HBSS with exponential bias function as follows:

$$p(r) = (1 - \beta) \beta^{(r-1)} \quad r \in \{1, \dots, \infty\} \quad (12)$$

$$= \left(\frac{1 - \beta}{\beta} \right) \beta^r \quad (13)$$

$$= \left(\frac{\beta}{1 - \beta} \right)^{-1} \beta^r \quad (14)$$

Based on the sum of an infinite geometric sequence $\sum_{k=0}^{\infty} ab^k = \frac{a}{1-b}$ for $|b| < 1$ and $a = b$:

$$= \left(\sum_{k=0}^{\infty} \beta^{(k+1)} \right)^{-1} \beta^r \quad (15)$$

$$= \left(\sum_{k=1}^{\infty} \beta^k \right)^{-1} \beta^r \quad (16)$$

which enables the blinking algorithm's probability function (Eq. 12) to be expressed as HBSS (Eq. 11) with exponential bias function ($bias(r) = \beta^r$) when there exists an infinite number of options ($R = \infty$):

$$p(r) = \left(\sum_{k=1}^R bias(k) \right)^{-1} bias(r) \quad \begin{cases} r \in \{1, \dots, R\} \\ R = \infty \\ bias(r) = \beta^r = \left(\frac{1}{\beta} \right)^{-r} \end{cases} \quad (17)$$

The condition $R = \infty$ implies HBSS bias weights should be divided by the infinite sum $\sum_{k=1}^{\infty} bias(k)$, a value approximated by the partial sum for strong bias functions. For example, HBSS probabilities with $bias(r) = e^{-r}$ are equal to those obtained by the blinking algorithm when $\beta = \frac{1}{e}$, while the blink rate $\beta = 0.01$ (applied in Section 5.3) is equal to HBSS with $bias(r) = 100^{-r}$. These results are presented in Table 10.

Table 10 Comparison of rank selection probabilities for HBSS and Blinks.

r	HBSS: $bias(r)$		r	Blinks: β	
	e^{-r}	100^{-r}		$\frac{1}{e}$	0.010
1	0.632	0.990	1	0.632	0.990
2	0.233	0.010	2	0.233	0.010
3	0.086	0.000	3	0.086	0.000
4	0.031	0.000	4	0.031	0.000
5	0.012	0.000	5	0.012	0.000
6-30	0.006	0.000	6- ∞	0.006	0.000

Appendix B: Detailed results for CVRP

Table 11 Aggregated computational results (Small).

Instance		BKS			ILS-SP (50 runs)			UHGS (50 runs)			SISRs (50 runs)		
#	Name	Dist	NV	Opt	Avg	Best	T	Avg	Best	T	Avg	Best	T
1	X-n101-k25	27591	26	✓	27591.0	27591	0.1	27591.0	27591	1.4	27591.0	27591	0.8
2	X-n106-k14	26362	14	✓	26375.9	26362	2.0	26381.8	26378	4.0	26381.5	26362	1.3
3	X-n110-k13	14971	13	✓	14971.0	14971	0.2	14971.0	14971	1.6	14971.1	14971	1.0
4	X-n115-k10	12747	10	✓	12747.0	12747	0.2	12747.0	12747	1.8	12747.0	12747	0.2
5	X-n120-k6	13332	6	✓	13337.6	13332	1.7	13332.0	13332	2.3	13332.0	13332	1.6
6	X-n125-k30	55539	30	✓	55673.8	55539	1.4	55542.1	55539	2.7	55556.3	55542	3.1
7	X-n129-k18	28940	18	✓	28998.0	28948	1.9	28948.5	28940	2.7	28948.8	28940	1.5
8	X-n134-k13	10916	13	✓	10947.4	10916	2.1	10934.9	10916	3.3	10940.1	10916	2.8
9	X-n139-k10	13590	10	✓	13603.1	13590	1.6	13590.0	13590	2.3	13595.4	13590	2.0
10	X-n143-k7	15700	7	✓	15745.2	15726	1.6	15700.2	15700	3.1	15705.8	15700	2.1
11	X-n148-k46	43448	47	✓	43452.1	43448	0.8	43448.0	43448	3.2	43469.2	43448	2.8
12	X-n153-k22	21220	23	✓	21400.0	21340	0.5	21226.3	21220	5.5	21229.4	21220	5.6
13	X-n157-k13	16876	13	✓	16876.0	16876	0.8	16876.0	16876	3.2	16878.6	16876	3.7
14	X-n162-k11	14138	11	✓	14160.1	14138	0.5	14141.3	14138	3.3	14157.1	14138	3.4
15	X-n167-k10	20557	10	✓	20608.7	20562	0.9	20563.2	20557	3.7	20560.8	20557	3.2
16	X-n172-k51	45607	53	✓	45616.1	45607	0.6	45607.0	45607	3.8	45619.2	45607	5.3
17	X-n176-k26	47812	26	✓	48249.8	48140	1.1	47957.2	47812	7.6	47849.6	47812	5.2
18	X-n181-k23	25569	23	✓	25571.5	25569	1.6	25591.1	25569	6.3	25579.8	25569	5.5
19	X-n186-k15	24145	15	✓	24186.0	24145	1.7	24147.2	24145	5.9	24178.4	24149	4.0
20	X-n190-k8	16980	8	✓	17143.1	17085	2.1	16987.9	16980	12.1	16984.9	16980	9.1
21	X-n195-k51	44225	53	✓	44234.3	44225	0.9	44244.1	44225	6.1	44298.5	44241	6.1
22	X-n200-k36	58578	36	✓	58697.2	58626	7.5	58626.4	58578	8.0	58636.1	58578	6.7
23	X-n204-k19	19565	19	✓	19625.2	19570	1.1	19571.5	19565	5.3	19662.3	19565	4.9
24	X-n209-k16	30656	16	✓	30765.4	30667	3.8	30680.4	30656	8.6	30669.4	30656	6.0
25	X-n214-k11	10856	11		11126.9	10985	2.3	10877.4	10856	10.2	10908.6	10873	8.7
26	X-n219-k73	117595	73	✓	117595.0	117595	0.8	117604.9	117595	7.7	117650.4	117595	8.0
27	X-n223-k34	40437	34	✓	40533.5	40471	8.5	40499.0	40437	8.3	40529.9	40448	7.6
28	X-n228-k23	25742	23	✓	25795.8	25743	2.4	25779.3	25742	9.8	25790.9	25744	10.5
29	X-n233-k16	19230	17		19336.7	19266	3.0	19288.4	19230	6.8	19269.7	19232	8.1
30	X-n237-k14	27042	14	✓	27078.8	27042	3.5	27067.3	27042	8.9	27089.7	27042	7.2
31	X-n242-k48	82751	48	✓	82874.2	82774	17.8	82948.7	82804	12.4	82884.4	82775	9.9
32	X-n247-k47	37274	51	✓	37507.2	37289	2.1	37284.4	37274	20.4	37323.2	37274	18.4

Table 12 Aggregated computational results (Medium).

Instance		BKS			ILS-SP (50 runs)			UHGS (50 runs)			SISRs (50 runs)		
#	Name	Dist	NV	Opt	Avg	Best	T	Avg	Best	T	Avg	Best	T
33	X-n251-k28	38684	28	✓	38840.0	38727	10.8	38796.4	38699	11.7	38791.0	38687	9.8
34	X-n256-k16	18880	17	✓	18883.9	18880	2.0	18880.0	18880	6.5	18888.9	18880	11.5
35	X-n261-k13	26558	13		26869.0	26706	6.7	26629.6	26558	12.7	26642.3	26558	11.8
36	X-n266-k58	75478	58	✓	75563.3	75478	10.0	75759.3	75517	21.4	75617.8	75478	10.8
37	X-n270-k35	35291	36	✓	35363.4	35324	9.1	35367.2	35303	11.2	35362.2	35323	11.4
38	X-n275-k28	21245	28	✓	21256.0	21245	3.6	21280.6	21245	12.0	21268.6	21245	13.3
39	X-n280-k17	33503	17		33769.4	33624	9.6	33605.8	33505	19.1	33628.1	33529	17.7
40	X-n284-k15	20226	15		20448.5	20295	8.6	20286.4	20227	19.9	20286.6	20240	15.3
41	X-n289-k60	95151	61	✓	95450.6	95315	16.1	95469.5	95244	21.3	95352.2	95233	14.3
42	X-n294-k50	47167	51		47254.7	47190	12.4	47259.0	47171	14.7	47274.5	47210	14.7
43	X-n298-k31	34231	31	✓	34356.0	34239	6.9	34292.1	34231	10.9	34276.0	34234	14.5
44	X-n303-k21	21744	21		21895.8	21812	14.2	21850.9	21748	17.3	21776.5	21751	17.3
45	X-n308-k13	25859	13		26101.1	25901	9.5	25895.4	25859	15.3	26207.7	25931	25.7
46	X-n313-k71	94044	72		94297.3	94192	17.5	94265.2	94093	22.4	94182.4	94063	18.9
47	X-n317-k53	78355	53	✓	78356.0	78355	8.6	78387.8	78355	22.4	78392.4	78355	22.0
48	X-n322-k28	29848	28		29991.3	29877	14.7	29956.1	29870	15.2	29927.6	29849	16.9
49	X-n327-k20	27546	20		27812.4	27599	19.1	27628.2	27564	18.2	27631.4	27608	21.6
50	X-n331-k15	31102	15	✓	31235.5	31105	15.7	31159.6	31103	24.4	31128.2	31122	20.4
51	X-n336-k84	139135	86		139461.0	139197	21.4	139534.9	139210	38.0	139373.4	139209	22.8
52	X-n344-k43	42068	43		42284.0	42146	22.6	42208.8	42099	21.7	42158.5	42079	21.5
53	X-n351-k40	25928	41		26150.3	26021	25.2	26014.0	25946	33.7	25982.1	25938	26.5
54	X-n359-k29	51505	29		52076.5	51706	48.9	51721.7	51509	34.9	51577.8	51505	23.1
55	X-n367-k17	22814	17		23003.2	22902	13.1	22838.4	22814	22.0	22833.4	22814	36.1
56	X-n376-k94	147713	94	✓	147713.0	147713	7.1	147750.2	147717	28.3	147783.6	147721	32.0
57	X-n384-k52	65943	53		66372.5	66116	34.5	66270.2	66081	40.2	66107.4	65963	25.9
58	X-n393-k38	38269	38		38457.4	38298	20.8	38374.9	38269	28.6	38394.1	38331	30.4
59	X-n401-k29	66187	29		66715.1	66453	60.4	66365.4	66243	49.5	66248.5	66189	38.0
60	X-n411-k19	19718	19		19954.9	19792	23.8	19743.8	19718	34.7	19768.5	19731	58.4
61	X-n420-k130	107798	130	✓	107838.0	107798	22.2	107924.1	107798	53.2	107879.2	107817	47.9
62	X-n429-k61	65483	62		65746.6	65563	38.2	65648.5	65501	41.5	65593.6	65485	35.0
63	X-n439-k37	36391	37	✓	36441.6	36395	39.6	36451.1	36395	34.5	36473.8	36426	42.1
64	X-n449-k29	55269	29		56204.9	55761	59.9	55553.1	55378	64.9	55411.2	55272	38.0
65	X-n459-k26	24173	26		24462.4	24209	60.6	24272.6	24181	42.8	24242.2	24175	56.5
66	X-n469-k138	221909	140		222182.0	221909	36.3	222617.1	222070	86.7	222227.1	221984	48.0
67	X-n480-k70	89458	70		89871.2	89694	50.4	89760.1	89535	67.0	89559.2	89458	50.5
68	X-n491-k59	66510	59		67226.7	66965	52.2	66898.0	66633	71.9	66645.5	66517	51.4

Table 13 Aggregated computational results (Large).

Instance		BKS			ILS-SP (50 runs)			UHGS (50 runs)			SISRs (50 runs)		
#	Name	Dist	NV	Opt	Avg	Best	T	Avg	Best	T	Avg	Best	T
69	X-n502-k39	69230	39		69346.8	69284	80.8	69328.8	69253	63.6	69274.7	69243	60.9
70	X-n513-k21	24201	21		24434.0	24332	35.0	24296.6	24201	33.1	24292.1	24238	77.1
71	X-n524-k137	154594	155	✓	155005.0	154709	27.3	154979.5	154774	80.7	154807.2	154651	151.4
72	X-n536-k96	94988	96		95700.7	95524	62.1	95330.6	95122	107.5	95173.2	95006	74.7
73	X-n548-k50	86701	50		86874.1	86710	64.0	86998.5	86822	84.2	86798.0	86710	64.5
74	X-n561-k42	42722	42		43131.3	42952	68.9	42866.4	42756	60.6	42868.1	42774	73.8
75	X-n573-k30	50719	30		51173.0	51092	112.0	50915.1	50780	188.2	50804.6	50737	113.0
76	X-n586-k159	190423	159		190919.0	190612	78.5	190838.0	190543	175.3	190600.7	190484	86.3
77	X-n599-k92	108490	93		109384.0	109056	73.0	109064.2	108813	125.9	108688.6	108548	75.4
78	X-n613-k62	59556	62		60444.2	60229	74.8	59960.0	59778	117.3	59731.3	59585	88.1
79	X-n627-k43	62210	43		62905.6	62783	162.7	62524.1	62366	239.7	62317.1	62219	89.3
80	X-n641-k35	63737	35		64606.1	64462	140.4	64192.0	63839	158.8	63850.3	63750	92.5
81	X-n655-k131	106780	131	✓	106782.0	106780	47.2	106899.1	106829	150.5	106844.6	106813	109.6
82	X-n670-k126	146451	133		147676.0	147045	61.2	147222.7	146705	264.1	146720.4	146451	198.9
83	X-n685-k75	68261	75		68988.2	68646	73.8	68654.1	68425	156.7	68369.0	68271	135.1
84	X-n701-k44	81934	44		83042.2	82888	210.1	82487.4	82293	253.2	82065.4	81974	122.5
85	X-n716-k35	43414	35		44171.6	44021	225.8	43641.4	43525	264.3	43483.8	43426	158.3
86	X-n733-k159	136250	160		137045.0	136832	111.6	136587.6	136366	244.5	136389.3	136255	143.2
87	X-n749-k98	77365	98		78275.9	77952	127.2	77864.9	77715	313.9	77509.2	77380	146.3
88	X-n766-k71	114525	71		115738.0	115443	242.1	115147.9	114683	383.0	114761.1	114590	174.4
89	X-n783-k48	72445	48		73722.9	73447	235.5	73009.6	72781	269.7	72660.7	72492	170.2
90	X-n801-k40	73331	40		74005.7	73830	432.6	73731.0	73587	289.2	73436.7	73347	137.1
91	X-n819-k171	158267	172		159425.0	159164	148.9	158899.3	158611	374.3	158423.0	158305	172.5
92	X-n837-k142	193813	142		195027.0	194804	173.2	194476.5	194266	463.4	193976.9	193824	166.8
93	X-n856-k95	89007	95		89277.6	89060	153.7	89238.7	89118	288.4	89131.3	89050	160.0
94	X-n876-k59	99331	59		100417.0	100177	409.3	99884.1	99715	495.4	99483.2	99388	217.4
95	X-n895-k37	53946	37		54958.5	54713	410.2	54439.8	54172	321.9	54085.8	53993	212.5
96	X-n916-k207	329247	207		330948.0	330639	226.1	330198.3	329836	560.8	329509.5	329299	215.3
97	X-n936-k151	132926	158		134530.0	133592	202.5	133512.9	133140	531.5	133117.3	133014	412.7
98	X-n957-k87	85482	87		85936.6	85697	311.2	85822.6	85672	432.9	85620.0	85546	202.4
99	X-n979-k58	119008	58		120253.0	119994	687.2	119502.1	119194	554.0	119120.4	119065	276.6
100	X-n1001-k43	72404	43		73985.4	73776	792.8	72956.0	72742	549.0	72528.1	72415	284.3

Appendix C: Detailed results for CVRP variants

Detailed results of 10 runs per instance are summarized in Tables 14 to 20. The instance name and best known solution (*BKS*) is provided for each problem in addition to, when applicable, the following problem characteristics: the number of customers n , vehicle capacity Q , maximum tour duration D , service time e (if equal for all customers), number of depots G , number of vehicles m (in each depot), number of linehaul l and backhaul b customers. Calculation times (Time) are reported in seconds. Best results are marked gray and new best results obtained by SISRs are underlined.

MDVRP (Results: Table 14)

The benchmark set employed for MDVRP was introduced by Cordeau, Gendreau, and Laporte (1997) with problem sizes varying between 48 and 360. The set is subdivided into three blocks. The first block of problems have no tour duration limits and therefore service time for each customer is zero. Route duration for the second block is limited, while customers have no service time. The third block of problems specify a route duration limit and an individual service time for each customer. Results are compared against those achieved by ALNS 50K (Pisinger and R pke 2007), ILS-RVND-SP (Subramanian, Uchoa, and Ochi 2013) and HGSADC (Vidal et al. 2012).

VRPB (Results: Table 15)

Goetschalckx and Jacobs-Blecha (1989) introduced a benchmark set for VRPB which contains problems containing a small number of customers, ranging from 25 to 150. Results are compared against those attained by ACS (Gajpal and Abad 2009), APA (Zachariadis and Kiranoudis 2011) and UHGS (Vidal et al. 2014).

VRPMDP and VRPSDP (Results: Table 16 and 17)

Benchmark instances employed for both problems are introduced by Salhi and Nagy (1999) and were reproduced using the instructions supplied. The authors based their single depot instances on those introduced by Christofides, Mingozzi, and Toth (1979) which have problem sizes from 50 to 199 customers. The problems are divided into two groups: the first with and the second without tour duration and service time constraints. Results are compared against those produced by ILS-RVND-SP (Subramanian, Uchoa, and Ochi 2013) and UHGS (Vidal et al. 2014).

MDVRPMDP (Results: Table 18)

The benchmark introduced by Salhi and Nagy (1999) was employed and reproduced using their instructions. This benchmark set is based on the Multi-Depot instances introduced by Gillett and Johnson (1976) with the number of customers ranging from 50 to 249 and the number of depots from 2 to 5. MDVRPMDP results are compared against those achieved by LNS (6R no learning) (R pke and Pisinger 2006b) and ILS-RVND-SP (Subramanian, Uchoa, and Ochi 2013).

OVRP (Results: Table 19)

Three benchmark sets were employed for OVRP: Christofides, Mingozzi, and Toth (1979), Fisher (1994) and Golden et al. (1998) with the number of customers ranging from 50 to 480 customers. Results are compared against those attained by ILS-RVND-SP (Subramanian, Uchoa, and Ochi 2013) and UHGS (Vidal et al. 2014).

CCVRP (Results: Table 20)

Benchmark sets introduced by Christofides, Mingozzi, and Toth (1979) and Golden et al. (1998) with problem sizes from 50 to 483 customers were employed. Results are compared against those produced by UHGS (Vidal et al. 2014).

The best results of five runs per instance are summerized in Tables 21 and 22. Results are divided in sets of problem sizes n 200, 400, 600, 800, and 1000. For each problem set, the cumulative number of vehicles (CNV), the cumulative travel distance (CTD) and calculation average calculation time (T) in minutes are reported. Best results are marked gray.

VRPTW (Results: Table 21)

The benchmark introduced by Gehring and Homberger (2001) was employed with the customers ranging from 200 to 1000. The number of iterations is set to 25×10^6 , 50×10^6 , 75×10^6 , 100×10^6 and 125×10^6 for small to large-scale instances. The best of five runs per instance are compared against those of RTI (Repoussis, Tarantilis, and Ioannou 2009), NB-100 and NB-f(n) (Nagata, Bräysy, and Dullaert 2010) and HGSADC (Vidal et al. 2013).

PDPTW (Results: Table 22)

The Benchmark introduced by *lilim2001* was employed with problem sizes varying between 200 and 1000. The number of iterations is set to 6×10^6 , 12×10^6 , 18×10^6 , 24×10^6 and 30×10^6 for small to large-scale instances. The best result of five runs per instance are compared against those attained by BH (Bent and Hentenryck 2006), RP (Røpke and Pisinger 2006a), NK (Nagata and Kobayashi 2010) and CLS (Curtois et al. 2018)

Table 14 MDVRP.

Instance	Characteristics					BKS	ALNS 50K		HGSADC		ILS-RVND-SP			SISRs		
	n	Q	D	m	G		Avg.	Time	Avg.	Time	Best	Avg.	Time	Best	Avg.	Time
p01	50	80	-	4	4	576.87	29	576.87	13.8	576.87	576.87	576.87	2.80	576.87	576.87	0.17
p02	50	160	-	2	4	473.53	28	473.53	12.6	473.53	473.53	473.53	2.27	473.53	473.53	0.12
p03	75	140	-	3	5	641.19	64	641.19	25.8	641.19	641.19	641.19	7.25	641.19	641.19	24.27
p12	80	60	-	5	2	1318.95	75	1318.95	31.2	1318.95	1318.95	1318.95	6.14	1318.95	1318.95	0.12
p04	100	100	-	8	2	1001.04	88	1001.23	116.4	1001.23	1001.04	1001.04	51.76	1001.04	1001.16	184.60
p05	100	200	-	5	2	752.34	120	750.03	63.6	750.03	750.03	750.21	31.54	750.03	750.19	197.51
p06	100	100	-	6	3	876.50	93	876.50	68.4	876.50	876.50	876.50	25.70	876.50	876.50	127.24
p07	100	100	-	4	4	881.97	88	889.36	93.0	884.43	881.97	881.97	21.88	881.97	881.97	141.20
p15	160	60	-	5	4	2505.42	253	2519.64	115.2	2505.42	2505.42	2505.42	48.59	2505.42	2505.42	3.00
p18	240	60	-	5	6	3702.85	419	3736.53	271.2	3702.85	3702.85	3702.85	1019.76	3702.85	3702.85	138.09
p21	360	60	-	5	9	5474.84	582	5501.58	600.0	5476.41	5474.84	5474.84	2544.57	5474.84	5476.61	688.68
p13	80	60	2000	5	2	1318.95	60	1318.95	34.2	1318.95	1318.95	1318.95	3.06	1318.95	1318.95	0.75
p14	80	60	1800	5	2	1360.12	58	1360.12	33.0	1360.12	1360.12	1360.12	19.11	1360.12	1360.12	0.05
p16	160	60	2000	5	4	2572.23	188	2573.95	118.2	2572.23	2572.23	2572.23	247.77	2572.23	2572.23	8.22
p17	160	60	1800	5	4	2709.09	179	2709.09	128.4	2709.09	2709.09	2710.21	1448.47	2709.09	2709.09	6.53
p19	240	60	2000	5	6	3827.06	315	3838.76	252.0	3827.06	3827.06	3827.55	1214.57	3827.06	3827.06	54.56
p20	240	60	1800	5	6	4058.07	300	4064.76	262.2	4058.07	4058.07	4058.07	544.80	4058.07	4058.07	29.06
p08	249	500	3100	14	2	4372.78	333	4421.03	600.0	4397.42	4379.46	4393.70	1244.57	4375.49	4381.91	983.77
p09	249	500	3100	12	3	3858.66	361	3892.50	570.0	3868.59	3859.54	3864.22	1431.88	3859.54	3870.47	930.73
p10	249	500	3100	8	4	3631.11	363	3666.85	589.2	3636.08	3631.37	3634.72	1422.66	3636.47	3647.09	1129.82
p11	249	500	3100	6	5	3546.06	357	3573.23	428.4	3548.25	3546.06	3546.15	1217.35	3546.06	3547.45	667.01
p22	360	60	2000	5	9	5702.16	462	5722.19	600.0	5702.16	5702.15	5705.84	846.01	5702.16	5703.70	220.20
p23	360	60	1800	5	9	6078.75	443	6092.66	600.0	6078.75	6078.75	6078.75	1019.15	6078.75	6078.75	101.16
pr01	48	200	5000	1	4	861.32	30	861.32	10.2	861.32	861.32	861.32	1.24	861.32	861.32	0.04
pr07	72	200	5000	1	6	1089.56	58	1089.56	20.4	1089.56	1089.56	1089.56	3.87	1089.56	1089.56	0.11
pr02	96	195	4800	2	4	1307.34	103	1308.17	45.6	1307.34	1307.34	1308.53	12.39	1307.34	1307.34	18.94
pr03	144	190	4600	3	4	1803.80	214	1810.66	114.6	1803.80	1803.81	1804.09	55.04	1803.80	1803.80	467.44
pr08	144	190	4750	2	6	1664.85	207	1675.74	123.0	1665.05	1664.85	1665.08	393.98	1664.85	1666.55	153.17
pr04	192	185	4400	4	4	2058.31	296	2073.16	313.2	2059.36	2058.31	2060.93	779.30	2058.31	2058.76	468.60
pr09	216	180	4500	3	6	2133.20	350	2144.84	366.0	2134.17	2133.20	2135.37	1070.41	2133.20	2135.05	787.91
pr05	240	180	4200	5	4	2331.20	372	2350.31	573.6	2340.29	2331.20	2338.12	1337.10	2331.20	2338.19	1124.96
pr06	288	175	4000	6	4	2676.30	465	2695.74	600.0	2681.93	2680.77	2685.23	2297.66	2680.74	2686.77	1808.75
pr10	288	170	4250	4	6	2868.26	455	2905.43	600.0	2886.59	2874.28	2882.41	3009.53	2867.46	2873.80	1417.51

Table 15 VRPB.

Instance	Characteristics					BKS	ACS		APA		UHGS			SISRs		
	n	Q	l	b	m		Best.	Avg.	Best	Avg.	Best	Avg.	Time	Best	Avg.	Time
A1	25	1550	20	5	8	229.89	229.89	229.89	229.89	229.89	229.89	229.89	6.60	229.89	229.89	0.05
A2	25	2550	20	5	5	180.12	180.12	180.12	180.12	180.12	180.12	180.12	7.20	180.12	180.12	0.07
A3	25	4050	20	5	4	163.41	163.41	163.41	163.41	163.41	163.41	163.41	7.80	163.41	163.41	0.09
A4	25	4050	20	5	3	155.80	155.80	155.80	155.80	155.80	155.80	155.80	9.60	155.80	155.80	0.04
B1	30	1600	20	10	7	239.08	239.08	239.08	239.08	239.08	239.08	239.08	8.40	239.08	239.08	0.06
B2	30	2600	20	10	5	198.05	198.05	198.05	198.05	198.05	198.05	198.05	9.00	198.05	198.05	0.03
B3	30	4000	20	10	3	169.37	169.37	169.37	169.37	169.37	169.37	169.37	10.80	169.37	169.37	0.02
C1	40	1800	20	20	7	250.56	250.56	250.56	250.56	250.56	250.56	250.56	13.20	250.56	250.56	0.38
C2	40	2600	20	20	5	215.02	215.02	215.02	215.02	215.02	215.02	215.02	14.40	215.02	215.02	0.13
C3	40	4150	20	20	5	199.35	199.35	199.35	199.35	199.35	199.35	199.35	13.80	199.35	199.35	0.07
C4	40	4150	20	20	4	195.37	195.37	195.37	195.37	195.37	195.37	195.37	14.40	195.37	195.37	0.04
D1	38	1700	30	8	12	322.53	322.53	322.53	322.53	322.53	322.53	322.53	10.80	322.53	322.53	0.58
D2	38	1700	30	8	11	316.71	316.71	316.71	316.71	316.71	316.71	316.71	10.20	316.71	316.71	0.23
D3	38	2750	30	8	7	239.48	239.48	239.48	239.48	239.48	239.48	239.48	11.40	239.48	239.48	0.10
D4	38	4075	30	8	5	205.83	205.83	205.83	205.83	205.83	205.83	205.83	14.40	205.83	205.83	0.06
E1	45	2650	30	15	7	238.88	238.88	238.88	238.88	238.88	238.88	238.88	16.20	238.88	238.88	0.05
E2	45	4300	30	15	4	212.26	212.26	212.26	212.26	212.26	212.26	212.26	19.20	212.26	212.26	0.12
E3	45	5225	30	15	4	206.66	206.66	206.66	206.66	206.66	206.66	206.66	21.60	206.66	206.66	0.23
F1	60	3000	30	30	6	263.17	263.17	263.17	263.27	263.17	263.17	263.17	22.80	263.17	263.17	1.04
F2	60	3000	30	30	7	265.21	265.21	265.21	265.66	265.21	265.21	265.21	23.40	265.21	265.21	0.71
F3	60	4400	30	30	5	241.12	241.12	241.48	241.12	241.12	241.12	241.12	29.40	241.12	241.12	6.38
F4	60	5500	30	30	4	233.86	233.86	233.86	234.60	233.86	233.86	233.86	32.40	233.86	233.86	0.12
G1	57	2700	45	12	10	306.31	306.31	306.54	?	?	?	?	?	306.31	306.31	2.05
G2	57	4300	45	12	6	245.44	245.44	245.44	245.44	245.44	245.44	245.44	22.80	245.44	245.44	0.14
G3	57	5300	45	12	5	229.51	229.51	229.51	229.97	229.51	229.51	229.51	25.80	229.51	229.51	1.81
G4	57	5300	45	12	6	232.52	232.52	232.52	232.52	232.52	232.52	232.52	27.00	232.52	232.52	6.20
G5	57	6400	45	12	5	221.73	221.73	221.73	222.87	221.73	221.73	221.73	27.60	221.73	221.73	3.36
G6	57	8000	45	12	4	213.46	213.46	213.46	214.38	213.46	213.46	213.46	32.40	213.46	213.46	0.31
H1	68	4000	45	23	6	268.93	268.93	269.00	268.93	270.06	268.93	268.93	37.20	268.93	268.93	0.80
H2	68	5100	45	23	5	253.37	253.37	253.37	253.91	253.37	253.37	253.37	34.20	253.37	253.37	0.35
H3	68	6100	45	23	4	247.45	247.45	247.45	247.45	247.45	247.45	247.45	38.40	247.45	247.45	0.65
H4	68	6100	45	23	5	250.22	250.22	250.22	251.09	250.22	250.22	250.22	35.40	250.22	250.22	1.07
H5	68	7100	45	23	4	246.12	246.12	246.12	246.12	246.12	246.12	246.12	37.20	246.12	246.12	1.58
H6	68	7100	45	23	5	249.14	249.14	249.14	250.06	249.14	249.14	249.14	35.40	249.14	249.14	1.45
I1	90	3000	45	45	10	350.25	350.25	350.40	350.25	351.08	350.25	350.37	53.40	350.25	350.25	52.81
I2	90	4000	45	45	7	309.94	309.94	310.32	309.94	309.98	309.94	309.94	51.00	309.94	309.94	1.04
I3	90	5700	45	45	5	294.51	294.51	294.84	294.51	294.79	294.51	294.51	59.40	294.51	294.51	0.71
I4	90	5700	45	45	6	295.99	295.99	296.13	295.99	297.91	295.99	295.99	55.20	295.99	295.99	1.04
I5	90	5700	45	45	7	301.24	301.24	301.83	301.24	303.49	301.24	301.24	49.20	301.24	301.24	10.45
J1	94	4400	75	19	10	335.01	335.01	335.12	335.01	335.78	335.01	335.01	49.80	335.01	335.01	17.47
J2	94	5600	75	19	8	310.42	310.42	310.42	310.42	312.51	310.42	310.42	50.40	310.42	310.42	31.41
J3	94	8200	75	19	6	279.22	279.22	279.34	279.22	280.43	279.22	279.22	55.80	279.22	279.22	5.60
J4	94	6600	75	19	7	296.53	296.53	296.58	296.53	298.32	296.53	296.53	67.20	296.53	296.53	65.91
K1	113	4100	75	38	10	394.07	394.07	396.14	394.07	397.38	394.07	394.35	79.80	394.07	394.09	163.42
K2	113	5200	75	38	8	362.13	362.13	362.56	362.13	365.46	362.13	362.13	84.00	362.13	362.13	53.27
K3	113	5200	75	38	9	365.69	365.69	366.71	365.69	369.44	365.69	365.69	78.00	365.69	365.69	48.18
K4	113	6200	75	38	7	348.95	348.95	350.32	348.95	349.72	348.95	348.95	77.40	348.95	348.95	32.20
L1	150	4400	75	75	10	417.90	417.90	420.06	417.90	421.68	417.90	418.16	236.40	417.90	417.90	234.84
L2	150	5000	75	75	8	401.23	401.23	401.36	401.23	405.20	401.23	401.23	177.60	401.23	401.23	172.41
L3	150	5000	75	75	9	402.68	402.68	404.32	402.68	405.76	402.68	402.68	163.80	402.68	402.68	161.56
L4	150	6000	75	75	7	384.64	384.64	384.83	384.64	388.14	384.64	384.64	141.00	384.64	384.64	29.11
L5	150	6000	75	75	8	387.56	387.56	390.33	387.57	390.46	387.56	387.56	163.20	387.56	387.56	56.01
M1	125	5200	100	25	11	398.59	398.59	399.12	398.59	400.50	398.59	398.66	94.80	398.59	398.59	244.32
M2	125	5200	100	25	10	396.92	396.92	398.16	396.92	401.91	396.92	396.93	143.40	396.92	397.24	246.44
M3	125	6200	100	25	9	375.70	375.70	377.81	375.70	378.07	375.70	375.93	162.00	375.70	375.76	272.26
M4	125	8000	100	25	7	348.14	348.14	348.46	348.14	352.03	348.14	348.20	103.20	348.14	348.31	190.71
N1	150	5700	100	50	11	408.10	408.10	408.17	408.10	411.72	408.10	408.10	163.20	408.10	408.10	269.33
N2	150	5700	100	50	10	408.07	408.07	408.25	408.07	412.31	408.07	408.13	153.00	408.07	408.07	268.53
N3	150	6600	100	50	9	394.34	394.34	394.70	394.34	398.76	394.34	394.94	147.60	394.34	394.34	177.09
N4	150	6600	100	50	10	394.79	394.79	394.87	394.79	396.16	394.79	395.13	142.20	394.79	394.79	173.45
N5	150	8500	100	50	7	373.48	373.48	374.12	373.48	376.90	373.48	373.55	187.20	373.48	373.48	173.56
N6	150	8500	100	50	8	373.76	373.76	374.79	373.76	379.78	373.76	373.76	205.20	373.76	373.76	207.83

Table 16 VRPMDP.

Instance	Characteristics				BKS		ILS-RVND-SP			UHGS			SISRs		
	n	Q	D	e			Best	Avg.	Time	Best	Avg.	Time	Best	Avg.	Time
CMT01H	50	160	-	-	3	465.02	465.02	465.03	2.07	465.02	465.02	41.4	465.02	465.02	17.31
CMT01Q	50	160	-	-	4	489.74	489.74	489.74	1.52	489.74	489.74	33.0	489.74	489.74	0.22
CMT01T	50	160	-	-	5	520.06	520.06	520.06	1.60	520.06	520.06	31.2	520.06	520.06	0.28
CMT02H	75	140	-	-	6	662.63	662.63	662.63	5.16	662.63	662.63	61.2	662.63	662.63	26.84
CMT02Q	75	140	-	-	8	731.26	731.26	731.40	8.03	731.26	731.45	87.6	731.26	731.62	111.10
CMT02T	75	140	-	-	9	782.77	782.77	782.77	7.56	782.77	782.77	49.8	782.77	782.77	78.28
CMT03H	100	200	-	-	4	700.94	700.94	700.94	17.65	700.94	700.94	116.4	700.94	700.94	286.63
CMT03Q	100	200	-	-	6	747.15	747.15	747.15	9.70	747.15	747.15	79.8	747.15	747.15	123.72
CMT03T	100	200	-	-	7	798.07	798.07	798.07	28.76	798.07	798.07	97.2	798.07	798.65	237.22
CMT12H	100	200	-	-	5	629.37	629.37	629.37	13.93	629.37	629.37	89.4	629.37	629.37	106.82
CMT12Q	100	200	-	-	7	729.25	729.25	729.25	17.37	729.25	729.25	91.8	729.25	729.54	134.39
CMT12T	100	200	-	-	9	787.52	787.52	787.52	6.79	787.52	787.52	57.0	787.52	787.52	25.08
CMT11H	120	200	-	-	4	818.05	818.05	818.06	63.18	818.05	818.05	271.8	818.05	818.05	317.07
CMT11Q	120	200	-	-	6	939.36	939.36	939.36	20.35	939.36	939.36	179.4	939.36	939.36	491.73
CMT11T	120	200	-	-	7	998.80	998.80	998.81	19.91	998.80	998.80	140.4	998.80	998.81	405.52
CMT04H	150	200	-	-	6	828.12	828.12	831.59	80.24	828.12	829.64	340.2	828.12	829.81	577.42
CMT04Q	150	200	-	-	9	913.93	915.27	915.27	58.92	915.27	915.46	250.8	915.27	915.31	684.49
CMT04T	150	200	-	-	11	990.39	990.39	990.39	50.42	990.39	990.39	150.0	990.42	992.53	730.89
CMT05H	199	200	-	-	9	978.74	978.74	978.74	1531.73	978.74	980.58	297.0	978.74	981.79	1169.66
CMT05Q	199	200	-	-	12	1104.87	1104.87	1105.79	1627.78	1104.87	1106.95	347.4	1106.39	1108.09	1268.31
CMT05T	199	200	-	-	15	1218.77	1218.77	1220.24	1802.81	1218.77	1220.53	357.6	1222.29	1222.96	1206.99
CMT06H	50	160	200	10	6	555.43	555.43	557.35	1.08	555.43	555.43	27.0	555.43	555.43	0.87
CMT06Q	50	160	200	10	6	555.43	555.43	557.15	1.08	555.43	555.43	27.0	555.43	555.43	0.89
CMT06T	50	160	200	10	6	555.43	555.43	556.64	1.15	555.43	555.43	27.6	555.43	555.43	0.53
CMT07H	75	140	160	10	11	900.12	900.54	900.84	4.47	900.12	900.59	61.8	900.12	900.33	166.67
CMT07Q	75	140	160	10	11	900.69	900.69	902.62	4.90	900.69	901.09	71.4	900.69	900.69	162.48
CMT07T	75	140	160	10	11	903.05	903.05	903.05	4.77	903.05	903.05	43.8	903.05	903.05	28.66
CMT08H	100	200	230	10	9	865.50	865.50	865.50	7.78	865.50	865.50	78.6	865.50	865.50	106.75
CMT08Q	100	200	230	10	9	865.50	865.50	865.50	7.50	865.50	865.50	79.2	865.50	865.50	97.49
CMT08T	100	200	230	10	9	865.54	865.54	865.54	7.18	865.54	865.54	78.6	865.54	865.54	123.07
CMT14H	100	200	1040	90	10	821.75	821.75	821.75	5.37	821.75	821.75	57.6	821.75	821.75	58.35
CMT14Q	100	200	1040	90	10	821.75	821.75	821.75	5.47	821.75	821.75	58.2	821.75	821.75	40.57
CMT14T	100	200	1040	90	10	826.77	826.77	826.77	6.30	826.77	826.77	69.0	826.77	826.77	56.85
CMT13H	120	200	720	50	11	1542.86	1542.86	1544.54	73.82	1542.86	1542.86	168.6	1542.86	1542.88	716.98
CMT13Q	120	200	720	50	11	1542.86	1542.86	1544.05	69.87	1542.86	1542.86	171.0	1542.86	1542.92	689.48
CMT13T	120	200	720	50	11	1542.86	1542.86	1544.11	73.59	1542.86	1542.95	171.6	1541.14	1542.69	683.08
CMT09H	150	200	200	10	14	1160.68	1160.68	1162.17	77.95	1160.68	1160.68	219.6	1160.68	1160.68	1223.96
CMT09Q	150	200	200	10	14	1161.24	1161.24	1161.69	80.64	1161.24	1161.32	207.6	1161.24	1161.24	927.37
CMT09T	150	200	200	10	14	1162.55	1162.55	1164.37	83.29	1162.55	1162.55	220.8	1162.55	1162.93	1134.01
CMT10H	199	200	200	10	17	1372.47	1372.52	1377.23	550.45	1372.47	1381.47	393.6	1372.20	1372.75	2422.02
CMT10Q	199	200	200	10	17	1374.18	1374.18	1379.47	537.66	1374.18	1378.41	474.6	1376.15	1379.40	2309.70
CMT10T	199	200	200	10	18	1381.04	1381.04	1388.17	501.65	1381.04	1390.82	329.4	1381.04	1387.80	1959.93

Table 17 **VRPSDP.**

Instance	Characteristics				BKS		ILS-RVND-SP			UHGS			SISRs		
	n	Q	D	e			Best	Avg.	Time	Best	Avg.	Time	Best	Avg.	Time
CMT01X	50	160	-	-	3	466.77	466.77	466.77	2.08	466.77	466.77	43.2	466.77	466.77	0.52
CMT01Y	50	160	-	-	3	466.77	466.77	466.77	1.97	466.77	466.77	42.6	466.77	466.77	0.44
CMT02X	75	140	-	-	6	684.21	684.21	684.78	12.79	684.21	684.43	79.2	684.21	684.21	107.93
CMT02Y	75	140	-	-	6	684.21	684.21	684.59	10.83	684.21	684.36	81.0	684.21	684.24	101.06
CMT03X	100	200	-	-	5	721.27	721.27	721.46	17.69	721.27	721.27	101.4	721.27	722.69	286.85
CMT03Y	100	200	-	-	5	721.27	721.27	721.50	17.61	721.27	721.27	107.4	721.27	722.33	324.51
CMT12X	100	200	-	-	5	662.22	662.22	663.44	9.07	662.22	662.22	104.4	662.22	662.22	127.09
CMT12Y	100	200	-	-	5	662.22	662.22	663.12	9.34	662.22	662.22	101.4	662.22	662.22	131.69
CMT11X	120	200	-	-	4	833.92	846.23	848.65	51.82	833.92	833.92	165.0	846.23	846.23	385.22
CMT11Y	120	200	-	-	4	833.92	846.23	848.74	48.63	833.92	833.92	159.6	833.92	845.00	364.32
CMT04X	150	200	-	-	7	852.46	852.46	853.02	98.03	852.46	852.65	249.6	852.46	852.50	682.37
CMT04Y	150	200	-	-	7	852.46	852.46	852.73	80.63	852.46	852.72	237.0	852.46	852.50	673.40
CMT05X	199	200	-	-	10	1029.25	1029.25	1029.52	1786.74	1029.25	1029.60	479.4	1029.25	1029.28	1211.06
CMT05Y	199	200	-	-	10	1029.25	1029.25	1029.25	1726.18	1029.25	1029.79	396.0	1029.25	1029.27	1276.72
CMT06X	50	160	200	10	6	555.43	555.43	557.35	1.04	-	-	-	555.43	555.43	0.51
CMT06Y	50	160	200	10	6	555.43	555.43	557.10	1.08	-	-	-	555.43	555.43	0.46
CMT07X	75	140	160	10	11	900.12	900.12	901.02	4.55	-	-	-	900.12	900.29	183.85
CMT07Y	75	140	160	10	11	900.12	900.12	901.08	4.87	-	-	-	900.12	900.20	194.80
CMT08X	100	200	230	10	9	865.50	865.50	865.50	7.36	-	-	-	865.50	865.50	96.30
CMT08Y	100	200	230	10	9	865.50	865.50	865.50	7.74	-	-	-	865.50	865.50	123.57
CMT14X	100	200	1040	90	10	821.75	821.75	821.75	5.42	-	-	-	821.75	821.75	46.98
CMT14Y	100	200	1040	90	10	821.75	821.75	821.75	5.48	-	-	-	821.75	821.75	53.17
CMT13X	120	200	720	50	11	1542.86	1542.86	1543.54	68.72	-	-	-	1542.86	1542.88	694.59
CMT13Y	120	200	720	50	11	1542.86	1542.86	1544.42	73.49	-	-	-	1542.86	1542.88	690.74
CMT09X	150	200	200	10	14	1160.68	1160.68	1161.77	64.43	-	-	-	1160.68	1160.68	1141.56
CMT09Y	150	200	200	10	14	1160.68	1160.68	1162.59	80.86	-	-	-	1160.68	1160.68	1197.02
CMT10X	199	200	200	10	17	1373.40	1373.40	1379.19	552.81	-	-	-	1374.00	1376.47	2367.67
CMT10Y	199	200	200	10	17	1373.40	1373.40	1377.03	547.39	-	-	-	1373.72	1375.97	2410.63

Table 18 MDVRPMDP.

Instance	Characteristics					BKS	LNS			ILS-RVND-SP			SISRs		
	n	Q	D	m	G		Best	Avg.	Time	Best	Avg.	Time	Best	Avg.	Time
GJ01H	50	80	-	4	4	499.12	499	499	?	499.12	499.12	?	499.12	499.12	1.42
GJ01Q	50	80	-	4	4	528.30	528	528	36	528.30	528.30	3.12	528.30	528.30	19.00
GJ01T	50	80	-	4	4	569.43	569	569	34	569.43	569.43	2.98	569.43	569.43	0.68
GJ02H	50	160	-	2	4	440.00	440	440	51	440.00	440.00	2.95	440.00	440.00	0.38
GJ02Q	50	160	-	2	4	449.72	450	451	43	449.72	449.72	2.60	449.72	449.72	0.84
GJ02T	50	160	-	2	4	464.13	464	464	37	464.13	464.13	2.50	464.13	464.13	0.20
GJ03H	75	140	-	3	5	579.45	581	583	81	579.45	579.45	9.54	579.45	579.45	31.12
GJ03Q	75	140	-	3	5	605.25	605	608	71	605.25	605.25	8.95	605.25	605.25	13.96
GJ03T	75	140	-	3	5	624.44	624	626	65	624.44	624.44	10.94	624.44	624.44	91.97
GJ04H	100	100	-	8	2	789.19	790	797	112	789.19	789.30	31.69	789.19	789.33	263.78
GJ04Q	100	100	-	8	2	874.78	875	876	94	874.78	874.79	41.89	874.78	874.78	221.18
GJ04T	100	100	-	8	2	962.25	962	969	85	962.25	962.65	37.95	962.25	964.20	192.07
GJ05H	100	200	-	5	2	676.81	678	680	168	676.81	676.91	22.24	676.81	676.81	272.28
GJ05Q	100	200	-	5	2	700.15	702	705	133	700.15	700.15	19.55	700.15	700.15	126.71
GJ05T	100	200	-	5	2	733.17	733	738	118	733.17	733.18	39.31	733.17	733.19	174.81
GJ06H	100	100	-	6	3	742.18	747	751	116	742.18	742.18	32.28	742.18	742.30	230.39
GJ06Q	100	100	-	6	3	793.85	794	800	100	793.85	793.87	25.00	793.85	793.85	151.85
GJ06T	100	100	-	6	3	850.82	851	853	90	850.82	850.82	27.19	850.82	850.82	170.51
GJ07H	100	100	-	4	4	732.73	733	734	117	732.73	732.73	24.66	732.73	732.73	114.70
GJ07Q	100	100	-	4	4	801.91	803	807	94	801.91	801.94	38.58	801.91	802.03	240.05
GJ07T	100	100	-	4	4	853.54	855	862	88	853.54	853.54	20.64	853.54	853.54	148.73
GJ08H	249	500	310	14	2	3320.39	3327	3373	581	3320.39	3342.91	1435.21	3303.39	3317.32	1295.56
GJ08Q	249	500	310	14	2	3745.18	3774	3810	479	3745.18	3769.01	1288.57	3732.56	3734.17	1083.53
GJ08T	249	500	310	14	2	4110.78	4134	4170	431	4110.78	4120.27	1272.63	4107.43	4115.18	1114.60
GJ09H	249	500	310	12	3	2990.92	3006	3028	646	2990.92	3005.52	1478.35	2988.24	2991.52	779.81
GJ09Q	249	500	310	12	3	3351.18	3355	3393	535	3351.18	3361.23	1362.18	3369.44	3377.57	489.14
GJ09T	249	500	310	12	3	3656.03	3677	3718	492	3656.03	3661.62	1316.84	3657.81	3658.69	1059.97
GJ10H	249	500	310	8	4	2894.71	2930	2963	644	2894.71	2905.23	1452.52	2890.84	2898.84	1249.03
GJ10Q	249	500	310	8	4	3220.79	3245	3267	513	3220.79	3226.79	1315.68	3221.10	3233.11	929.20
GJ10T	249	500	310	8	4	3470.70	3485	3524	472	3470.70	3477.99	1281.92	3478.11	3493.40	815.96
GJ11H	249	500	310	6	5	2842.79	2880	2905	609	2842.79	2845.71	1357.58	2840.21	2843.71	1041.71
GJ11Q	249	500	310	6	5	3138.64	3165	3192	511	3138.64	3143.33	1267.12	3143.85	3152.85	1440.65
GJ11T	249	500	310	6	5	3360.48	3390	3421	469	3360.48	3367.63	1181.56	3365.39	3370.03	967.63

Table 19 OVRP.

Instance	Characteristics				BKS			ILS-RVND-SP			UHGS			SISRs		
	n	Q	D	e				Best	Avg.	Time	Best	Avg.	Time	Best	Avg.	Time
CMT01	50	160	-	-	5	416.06		416.06	416.06	1.78	416.06	416.06	24.6	416.06	416.06	1.07
F11	71	30000	-	-	4	177.00		177.00	177.21	4.41	177.00	177.00	39.0	177.00	177.00	18.46
CMT02	75	140	-	-	10	567.14		567.14	567.14	6.44	567.14	568.15	30.6	567.14	567.14	36.14
CMT03	100	200	-	-	8	639.74		639.74	639.81	15.67	639.74	639.74	51.0	639.74	639.74	193.38
CMT12	100	200	-	-	10	534.24		534.24	534.24	5.74	534.24	534.24	36.6	534.24	534.24	114.24
CMT11	120	200	-	-	7	682.12		682.12	682.12	23.54	682.12	682.12	90.6	682.12	682.12	232.22
F12	134	2210	-	-	7	769.55		769.55	770.00	40.51	769.55	769.68	102.6	786.14	787.39	559.19
CMT04	150	200	-	-	12	733.13		733.13	733.13	27.77	733.13	733.13	103.8	733.13	733.13	549.90
CMT05	199	200	-	-	16	883.50		883.50	895.55	1579.45	884.08	890.15	247.8	884.01	886.26	492.73
CMT06	50	160	180	10	6	412.96		412.96	412.96	1.24	412.96	412.96	33.0	412.96	412.96	0.69
CMT07	75	140	144	10	10	583.19		583.19	582.07	7.29	583.19	584.59	46.2	583.19	583.19	53.04
CMT08	100	200	207	10	9	644.63		644.63	644.95	9.21	644.63	644.79	107.4	644.63	644.63	222.03
CMT14	100	200	936	90	11	591.87		591.87	591.87	22.76	591.87	591.87	102.0	591.87	591.87	138.92
CMT13	120	200	648	50	11	899.16		899.16	904.02	37.35	899.16	900.22	203.4	902.25	909.17	479.60
CMT09	150	200	180	10	13	757.07		757.91	759.38	38.93	757.07	760.75	310.8	757.07	757.07	543.42
CMT10	199	200	180	10	17	874.71		874.71	877.68	472.91	874.71	875.49	366.0	874.71	874.71	898.35
GWKC5	200	900	-	-	5	6018.52		6018.52	6018.52	1270.81	6018.52	6018.52	342.6	6018.52	6018.52	10.67
GWKC1	240	550	-	-	9	4543.00		4544.46	4551.74	1247.37	4543.00	4546.35	492.0	4546.94	4556.00	975.73
GWKC6	280	900	-	-	7	7704.59		7721.16	7728.77	2008.76	7704.59	7704.91	673.8	7705.97	7709.74	1035.73
GWKC2	320	700	-	-	10	7213.56		7215.48	7229.56	2663.24	7213.56	7218.41	896.4	7213.56	7216.62	2220.94
GWKC7	360	900	-	-	8	9127.70		9180.93	9205.01	5702.38	9127.70	9132.27	1140.6	9127.70	9138.40	1348.60
GWKC3	400	900	-	-	9	9750.63		9773.83	9784.52	5065.72	9750.63	9763.39	1359.6	9750.63	9757.72	2702.40
GWKC8	440	900	-	-	10	10289.70		10326.57	10342.10	6140.55	10289.70	10316.60	1585.8	10293.87	10307.55	3760.46
GWKC4	480	1000	-	-	10	12380.66		12389.43	12393.40	6653.41	12380.66	12387.82	1584.6	12372.97	12390.73	3976.79

Table 20 CCVRP.

Instance	Characteristics			BKS	UHGS			SISRs		
	n	Q	m		Best	Avg.	Time	Best	Avg.	Time
CMT1	50	160	5	2230.35	2230.35	2230.35	26.4	2230.35	2230.35	2.64
CMT2	75	140	10	2391.63	2391.63	2394.00	42.0	2391.63	2391.63	8.67
CMT3	100	200	8	4045.42	4045.42	4045.42	55.8	4045.42	4045.42	21.04
CMT4	150	200	12	4987.52	4987.52	4987.52	110.4	4987.52	4987.52	61.23
CMT5	199	200	17	5806.02	5806.02	5809.94	241.2	5806.02	5809.11	144.64
CMT6	120	200	7	7314.55	7314.55	7314.55	81.0	7314.55	7314.55	32.41
CMT7	100	200	10	3558.92	3558.93	3558.93	38.4	3558.92	3558.92	32.72
GWKC1	240	550	9	54739.85	54739.85	54742.20	444.0	54739.84	54742.20	399.42
GWKC2	320	700	10	100560.16	100560.16	100562.52	660.0	100560.17	100575.37	1055.56
GWKC3	400	900	10	170923.53	170923.53	170964.42	1437.0	170923.55	170958.03	1702.85
GWKC4	480	1000	10	261993.33	261993.33	262044.19	1568.4	261993.34	262031.30	1929.01
GWKC5	200	900	5	114163.63	114163.63	114163.63	417.0	114163.63	114170.48	263.15
GWKC6	280	900	7	140430.08	140430.08	140430.08	759.0	140430.08	140430.78	699.12
GWKC7	360	900	8	178880.44	178880.44	178976.20	1599.6	179292.14	179487.17	451.20
GWKC8	440	900	10	193659.14	193659.14	193683.21	1564.2	193661.16	193738.98	1585.11
GWKC9	255	1000	14	4722.06	4722.06	4724.01	418.2	4724.36	4725.23	560.01
GWKC10	323	1000	16	6713.26	6713.26	6720.04	618.0	6713.12	6715.38	1069.13
GWKC11	399	1000	18	9214.07	9219.42	9222.92	876.6	9212.90	9214.70	1781.35
GWKC12	483	1000	19	12500.52	12500.52	12516.98	1719.6	12488.05	12492.29	1501.60
GWKC13	252	1000	26	3627.45	3627.45	3632.63	429.6	3628.21	3630.74	270.23
GWKC14	320	1000	29	5187.56	5187.56	5206.53	975.0	5189.71	5198.27	323.61
GWKC15	396	1000	33	7005.47	7005.47	7015.51	1122.0	7001.25	7011.07	633.32
GWKC16	480	1000	37	9239.10	9239.10	9247.68	1636.8	9236.44	9241.93	1390.13
GWKC17	240	200	22	3060.14	3060.14	3061.28	363.6	3059.82	3061.45	273.71
GWKC18	300	200	27	4199.43	4199.43	4211.80	698.4	4199.42	4204.36	287.94
GWKC19	360	200	33	5496.39	5496.39	5502.59	1023.6	5500.05	5508.30	661.72
GWKC20	420	200	38	7184.19	7184.19	7188.59	1309.2	7186.75	7190.16	807.04

Table 21 VRPTW.

Inst.	n	RTI	NB-100	NB-f(n)	HGSADC	SISRs
		Best 3	1 run	Best 5	Best 5	Best 5
R1	200	18.2 3640.11	18.2 3615.15	18.2 3612.36	18.2 3613.16	18.2 3619.20
R2	200	4.0 2941.99	4.0 2930.04	4.0 2929.41	4.0 2929.41	4.0 2939.40
C1	200	18.9 2721.90	18.9 2718.44	18.9 2718.41	18.9 2718.41	18.9 2718.50
C2	200	6.0 1833.36	6.0 1831.64	6.0 1831.64	6.0 1831.59	6.0 1837.80
RC1	200	18.0 3224.63	18.0 3182.48	18.0 3178.68	18.0 3180.48	18.0 3184.30
RC2	200	4.3 2554.33	4.3 2536.54	4.3 2536.22	4.3 2536.20	4.3 2545.60
CNV		694	694	694	694	694
CTD		169163	168143	168067	168092	168447
T		90	4.7	5x4.1	5x8.40	5x11.0
R1	400	36.4 8514.11	36.4 8413.23	36.4 8403.24	36.4 8402.57	36.4 8399.00
R2	400	8.0 6258.82	8.0 6149.49	8.0 6148.57	8.0 6152.92	8.0 6196.20
C1	400	37.6 7273.90	37.6 7179.71	37.6 7175.72	37.6 7170.47	37.6 7200.90
C2	400	11.7 3941.70	11.7 3898.02	11.7 3899.00	11.6 3952.95	11.8 3927.10
RC1	400	36.0 8088.46	36.0 7931.66	36.0 7922.23	36.0 7907.14	36.0 7906.60
RC2	400	8.4 5516.59	8.4 5293.74	8.4 5297.86	8.5 5215.21	8.4 5356.40
CNV		1381	1381	1381	1381	1382
CTD		395936	388548	388466	388013	389861
T		180	34.0	5x16.2	5x34.1	5x42.0
R1	600	54.5 18781.79	54.5 18194.38	54.5 18186.24	54.5 18023.18	54.5 17946.80
R2	600	11.0 12804.60	11.0 12319.75	11.0 12330.49	11.0 12352.38	11.0 12485.80
C1	600	57.3 14236.86	57.4 14054.70	57.4 14067.34	57.4 14058.46	57.4 14076.60
C2	600	17.4 7729.80	17.4 7601.94	17.4 7605.07	17.4 7594.41	17.4 7774.20
RC1	600	55.0 16767.72	55.0 16179.39	55.0 16183.95	55.0 16097.05	55.0 16018.50
RC2	600	11.4 11311.81	11.4 10591.87	11.4 10586.14	11.5 10511.86	11.4 10776.50
CNV		2066	2067	2067	2068	2067
CTD		816326	789420	789592	786373	790784
T		270	80.4	5x25.3	5x99.4	5x95.3
R1	800	72.8 32734.57	72.8 31486.74	72.8 31492.81	72.8 31311.38	72.8 31208.50
R2	800	15.0 20618.21	15.0 19873.04	15.0 19914.97	15.0 19933.39	15.0 20082.50
C1	800	75.2 25911.44	75.3 24990.42	75.2 25151.83	75.4 24876.38	74.9 25530.30
C2	800	23.4 11835.72	23.4 11438.52	23.4 11447.27	23.3 11475.05	23.4 11638.20
RC1	800	72.0 33795.61	72.0 31020.22	72.0 31278.28	72.0 29404.32	72.8 28665.70
RC2	800	15.5 17536.54	15.4 16438.90	15.4 16484.31	15.4 16495.82	15.4 16655.10
CNV		2739	2739	2738	2739	2743
CTD		1424321	1352478	1357695	1334963	1337802
T		360	126.8	5x27.6	5x215	5x216.6
R1	1000	91.9 51414.26	91.9 48287.98	91.9 48369.71	91.9 47759.66	91.9 47338.80
R2	1000	19.0 30804.79	19.0 28913.40	19.0 29003.42	19.0 29076.45	19.0 29246.40
C1	1000	94.2 43111.60	94.1 41683.29	94.1 41748.60	94.1 41572.86	93.9 41827.70
C2	1000	29.3 16810.22	29.1 16498.61	29.1 16534.36	28.8 16796.45	29.1 16733.70
RC1	1000	90.0 46753.61	90.0 44743.18	90.0 44860.60	90.0 44333.40	90.0 44150.80
RC2	1000	18.4 25588.52	18.3 23939.62	18.3 24055.31	18.2 24131.13	18.2 24338.60
CNV		3428	3424	3424	3420	3421
CTD		2144830	2040661	2045720	2036700	2036358
T		450	186.4	5x35.3	5x349	5x324.2

Table 22 PDPTW.

n	BH (10 runs)			RP (5 runs)			NK (5 runs)			CLS (All runs)			SISRs (5 runs)		
	CNV	CTD	Time	CNV	CTD	Time	CNV	CTD	Time	CNV	CTD	Time	CNV	CTD	Time
200	614	180358	65.0	606	180931	4.4	606	179996	7.6	601	186159	15.0	600	185257	6.3
400	1188	423636	100.0	1158	422201	14.7	1158	418455	40.2	1142	447627	15.0	1133	444247	23.7
600	1718	879940	100.0	1679	863442	37.0	1670	851174	81.1	1643	935948	30.0	1629	906717	66.0
800	2245	1480767	135.0	2208	1432078	65.3	2189	1416710	227.8	2146	1551495	60.0	2119	1526559	140.1
1000	2759	2225190	135.0	2652	2137034	89.5	2659	2120760	275.1	2634	2310830	60.0	2577	2276009	226.6

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